

10.13.1.8 (ID 3608) Baseline Data Element 8

Location in the menu: **V2.16.1.8**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.9 (ID 3609) Baseline Data Element 9

Location in the menu: **V2.16.1.9**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.10 (ID 3610) Baseline Data Element 10

Location in the menu: **V2.16.1.10**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.11 (ID 3622) Baseline Run Status

Location in the menu: **V2.16.1.11**

This monitoring value shows the baseline run status.

10.13.2 Stator Winding Monitor

10.13.2.1 (ID 3617) Current Unbalance

Location in the menu: **V2.16.2.1**

This monitoring value shows the current unbalance when condition-based monitoring is activated.

10.13.2.2 (ID 3620) Current Unbalance Threshold Value

Location in the menu: **V2.16.2.2**

This monitoring value shows the threshold value after interpolating.

10.13.2.3 (ID 3611) Current Warning S1 High Threshold

Location in the menu: **V2.16.2.3**

This monitoring value shows the warning S1 high threshold value.

10.13.2.4 (ID 3612) Current Warning S2 High Threshold

Location in the menu: **V2.16.2.4**

This monitoring value shows the warning S2 high threshold value.

10.13.2.5 (ID 3613) Current Alarm High Threshold

Location in the menu: **V2.16.2.5**

This monitoring value shows the alarm/fault high threshold value.

10.13.3 Vibration

10.13.3.1 (ID 3623) Vibration

Location in the menu: **V2.16.3.1**

This monitoring value shows the vibration when condition-based monitoring is activated.

10.13.3.2 (ID 3624) Vibration Threshold Value

Location in the menu: **V2.16.3.2**

This monitoring value shows the threshold value after interpolating.

10.13.3.3 (ID 3625) Vibration S1 High Threshold

Location in the menu: **V2.16.3.3**

This monitoring value shows the warning S1 high threshold value.

10.13.3.4 (ID 3626) Vibration S2 High Threshold

Location in the menu: **V2.16.3.4**

This monitoring value shows the warning S2 high threshold value.

10.13.3.5 (ID 3627) Vibration Alarm High Threshold

Location in the menu: **V2.16.3.5**

This monitoring value shows the alarm/fault high threshold value.

10.13.4 Load

10.13.4.1 (ID 3628) Load Threshold Value

Location in the menu: **V2.16.4.2**

This monitoring value shows the threshold value after interpolating.

10.13.4.2 (ID 3629) Load S1 High Threshold

Location in the menu: **V2.16.4.3**

This monitoring value shows the warning S1 high threshold value.

10.13.4.3 (ID 3630) Load S2 High Threshold

Location in the menu: **V2.16.4.4**

This monitoring value shows the warning S2 high threshold value.

10.13.4.4 (ID 3631) Load Alarm High Threshold

Location in the menu: **V2.16.4.5**

This monitoring value shows the alarm/fault high threshold value.

10.13.4.5 (ID 3632) Load S1 Low Threshold

Location in the menu: **V2.16.4.6**

This monitoring value shows the warning S1 low threshold value.

10.13.4.6 (ID 3633) Load S2 Low Threshold

Location in the menu: **V2.16.4.7**

This monitoring value shows the warning S2 low threshold value.

10.13.4.7 (ID 3634) Load Alarm Low Threshold

Location in the menu: **V2.16.4.8**

This monitoring value shows the alarm/fault low threshold value.

10.13.5 (ID 3619) Condition-based Monitoring Status

Location in the menu: **V2.16.5**

This monitoring value shows the fault status of the condition-based monitoring.

11 Parameter Descriptions

11.1 (ID 212) Application

Location in the menu: **P1.2**

Use this parameter to select the application configuration for the drive. The applications include preset application configurations, that is, sets of predefined parameters. The selection of the application makes the commissioning of the drive easy and reduces the manual work with the parameters.

⚠ WARNING ⚠

ACCIDENTAL START BECAUSE OF CHANGE OF SETTINGS

Running the Startup wizard, changing the application or the software can cause the I/O functions to change.

- Disconnect the motor from the drive if an accidental start can be dangerous.

When the value of this parameter changes, a parameter groups get their preset values. The value of this parameter can be changed when the starting up or commissioning the drive.

If this parameter is changed in the control panel, an application wizard starts and helps to set the basic parameters related to the application. The wizard does not start, when using the PC tool to change this parameter. For more data about the application wizards, see [4.1 Application Wizards](#).

These applications are available:

- 0 = Standard
- 1 = HVAC
- 3 = PID control
- 4 = Multi-ump (single drive)
- 5 = Multi-ump (multidrive)

N O T I C E

When the application is changed, the contents of the Quick Setup menu change.

11.2 Trend Curve

11.2.1 (ID 2368) Sampling Interval

Location in the menu: **P2.2.2**

Use this parameter to set the sampling interval.

11.2.2 (ID 2369) Channel 1 MIN

Location in the menu: **P2.2.3**

This parameter is used in scaling by default. Adjustments can be necessary.

11.2.3 (ID 2370) Channel 1 MAX

Location in the menu: **P2.2.4**

This parameter is used in scaling by default. Adjustments can be necessary.

11.2.4 (ID 2371) Channel 2 MIN

Location in the menu: **P2.2.5**

This parameter is used in scaling by default. Adjustments can be necessary.

11.2.5 (ID 2372) Channel 2 MAX

Location in the menu: **P2.2.6**

This parameter is used in scaling by default. Adjustments can be necessary.

11.2.6 (ID 2373) Autoscale

Location in the menu: **P2.2.7**

Use this parameter to set autoscaling on or off. If autoscaling is enabled, the signal is automatically scaled between the minimum and maximum values.

11.3 Motor Settings

11.3.1 Motor Nameplate Parameters

11.3.1.1 (ID 110) Motor Nominal Voltage

Location in the menu: **P3.1.1.1**

Find the value U_n on the nameplate of the motor. Find out whether the motor connection is Delta or Star.

11.3.1.2 (ID 111) Motor Nominal Frequency

Location in the menu: **P3.1.1.2**

Find the value f_n on the nameplate of the motor.

When this parameter changes, parameters **P3.1.4.2 Field Weakening Point Frequency** and **P3.1.4.3 Voltage at Field Weakening Point** start automatically. The 2 parameters have different values for each motor type. See [11.3.2.1 \(ID 650\) Motor Type](#).

11.3.1.3 (ID 112) Motor Nominal Speed

Location in the menu: **P3.1.1.3**

Find the value n_n on the nameplate of the motor.

11.3.1.4 (ID 113) Motor Nominal Current

Location in the menu: **P3.1.1.4**

Find the value I_n on the nameplate of the motor.

11.3.1.5 (ID 120) Motor Cos Phi

Location in the menu: **P3.1.1.5**

Find the value on the nameplate of the motor.

11.3.1.6 (ID 116) Motor Nominal Power

Location in the menu: **P3.1.1.6**

Find the value P_n on the nameplate of the motor.

11.3.2 Motor Control Parameters

11.3.2.1 (ID 650) Motor Type

Location in the menu: **P3.1.2.2**

Use this parameter to set the type of motor in the process.

Selection number	Selection name	Description
0	Induction motor (IM)	Make this selection if an induction motor is in use.
1	Permanent Magnet Motor (PM)	Make this selection if permanent magnet motor is in use.
2	Reluctance Motor	Make this selection if a reluctance motor is in use.

When the value of parameter **P3.1.2.2 Motor Type** is changed, the values of parameters **P3.1.4.2 Field Weakening Point Frequency** and **P3.1.4.3 Voltage at Field Weakening Point** change automatically. See the following table. The 2 parameters have different values for each motor type.

Parameter	Induction motor (IM)	Permanent magnet motor (PM)
P3.1.4.2 (Field Weakening Point Frequency)	Motor nominal frequency	Internally calculated
P3.1.4.3 (Voltage at Field Weakening Point)	100.0%	Internally calculated

11.3.2.2 (ID 601) Switching Frequency

Location in the menu: **P3.1.2.3**

Use this parameter to set the switching frequency of the AC drive.

If the switching frequency is increased, the capacity of the AC drive reduces. To reduce capacitive currents in the motor cable, when the cable is long, we recommend using a low switching frequency. To reduce the motor noise, use a high switching frequency.

11.3.2.3 (ID 631) Identification

Location in the menu: **P3.1.2.4**

Use this parameter to find the parameter values that are optimal for the operation of the drive.

The identification run calculates or measures the motor parameters that are necessary for a good control of the motor and speed.

The identification run helps to adjust the motor-specific and the drive-specific parameters. It is a tool for the commissioning and the servicing of the drive.

NOTICE

Before the identification run, set the motor nameplate parameters.

Selection number	Selection name	Description
0	No action	No identification requested.
1	Identification at standstill	The drive operates without speed when doing the identification run for the motor parameters. The motor receives current and voltage, but the frequency is zero. The U/f ratio and start magnetization parameters are identified.
2	Identification with rotation	The drive operates with speed when doing the identification run for the motor parameters. The U/f ratio, the magnetization current and start magnetization parameters are identified. To get accurate results, do this identification run with no load on the motor shaft.

To activate the Identification function, set the parameter P3.1.2.4 and give a start command. The start command must be given in 20 s. If there is no start command in that time, the identification run does not start. The parameter P3.1.2.4 is reset to the default value and an identification alarm shows.

To stop the identification run before it is completed, give a stop command. This resets the parameter to the default value. If the identification run is not completed, an identification alarm shows.

NOTICE

To start the drive after the identification, a new start command is necessary.

11.3.2.4 (ID 612) Magnetizing Current

Location in the menu: **P3.1.2.5**

Use this parameter to set the magnetizing current of the motor.

The magnetizing current (no-load current) of the motor identifies the values of the U/f parameters when they are given before the identification run. If the value is set to 0, the magnetizing current is calculated internally.

11.3.2.5 (ID 653) Motor Switch

Location in the menu: **P3.1.2.6**

Use this parameter to enable the Motor Switch function. Use the Motor Switch function, if the cable that connects the motor and the drive has a motor switch. The operation of the motor switch makes sure that the motor is isolated from the voltage source and does not start during the servicing.

To activate the function, set the parameter P3.1.2.6 to the value Enabled. The drive stops automatically when the motor switch is opened, and the drive starts automatically when the motor switch is closed. The drive does not trip when using the Motor switch function.

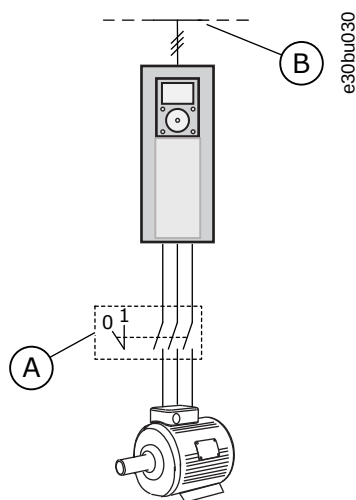


Illustration 35: The Motor Switch Between the Drive and the Motor

A	The motor switch
B	Mains

11.3.2.6 (ID 607) Overvoltage Control

Location in the menu: **P3.1.2.10**

Use this parameter to set the overvoltage controller out of operation.

The function is necessary when

- the supply voltage changes, for example, between -15% and +10%, and
- the controlled process does not have the tolerance for the changes that the undervoltage controller and the overvoltage controller make to the output frequency of the drive.

The overvoltage controller increases the output frequency of the drive

- to keep the DC link voltage in the permitted limits, and
- to make sure that the drive does not trip because of an overvoltage fault.

NOTICE

The drive can trip when the overvoltage and undervoltage controllers are disabled.

11.3.2.7 (ID 608) Undervoltage Control

Location in the menu: **P3.1.2.11**

Use this parameter to set the undervoltage controller out of operation.

The function is necessary when

- the supply voltage changes, for example, between -15% and +10%, and
- the controlled process does not have the tolerance for the changes that the undervoltage controller and the overvoltage controller make to the output frequency of the drive.

The undervoltage controller decreases the output frequency of the drive

- to get energy from the motor to keep the DC-link voltage at a minimum level when the voltage is near the lowest permitted limit, and
- to make sure that the drive does not trip because of an undervoltage fault.

N O T I C E

The drive can trip when the overvoltage and undervoltage controllers are disabled.

11.3.2.8 (ID 666) Energy Optimization

Location in the menu: **P3.1.2.12**

Use this parameter to enable the Energy Optimization function. To save energy and to lower the motor noise, the drive searches for the minimum motor current. Use this function, for example in fan and pump processes. Do not use the function with fast PID controlled processes.

11.3.2.9 (ID 659) Stator Voltage Adjust

Location in the menu: **P3.1.2.13**

Use this parameter to adjust the stator voltage in permanent magnet motors.

N O T I C E

The identification run sets a value for this parameter automatically. We recommend the identification run, if it is possible. Make the identification run with the parameter P3.1.2.4.

It is possible to use this parameter only when the parameter P3.1.2.2 Motor Type has the value PM motor. If induction motor is selected as the motor type, the value is automatically set to 100%, and the value cannot be changed.

When changing the value of P3.1.2.2 (Motor type) to PM Motor, the parameters P3.1.4.2 (Field Weakening Point Frequency) and P3.1.4.3 (Voltage at Field Weakening Point) increase automatically to be equal with output voltage of the drive. The set U/f ratio does not change. This is done to prevent the operation of the PM motor in the field weakening area. The nominal voltage of the PM motor is much lower than the full output voltage of the drive.

The nominal voltage of the PM motor agrees to the back EMF voltage of the motor at nominal frequency. But in a different motor manufacturer, it can be equal to, for example, the stator voltage at nominal load.

Stator Voltage Adjust helps to adjust the U/f curve of the drive near the back EMF curve. It is not necessary to change the values of many U/f curve parameters.

The parameter P3.1.2.13 gives the output voltage of the drive in percentage of the nominal voltage of the motor at the nominal frequency of the motor. Adjust the U/f curve of the drive above the back EMF curve. The motor current increases the more the U/f curve is different from the back EMF curve.

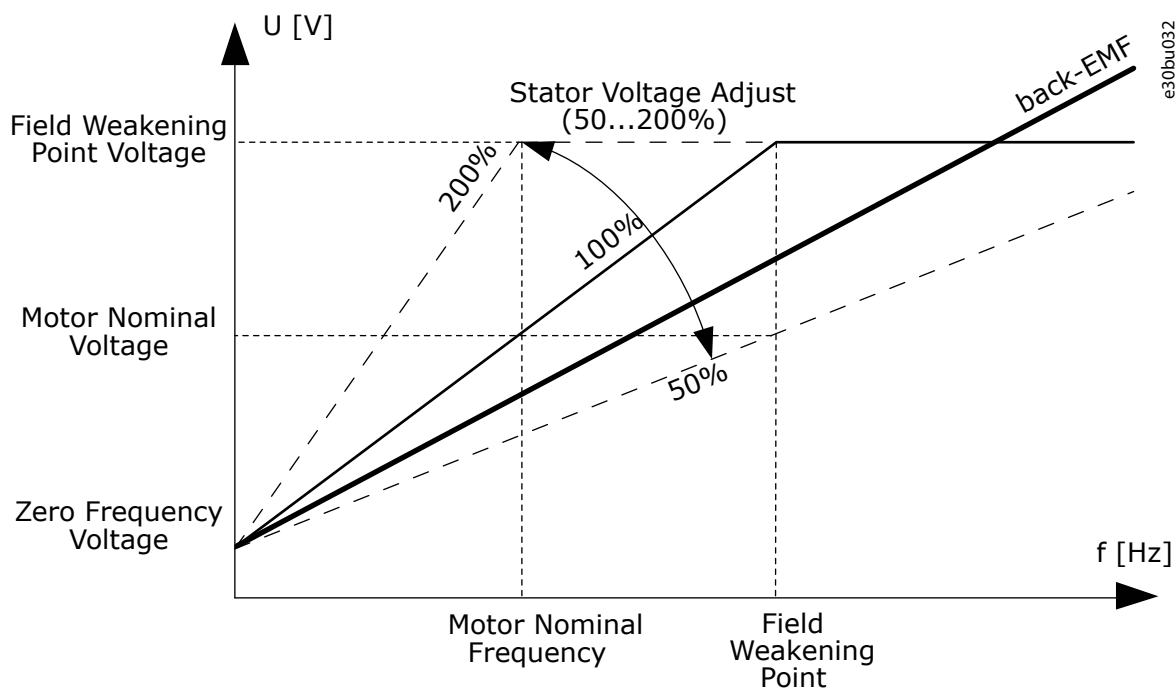


Illustration 36: The Stator Voltage Adjustment

11.3.2.10 (ID 662) Stator Resistance Voltage Drop

Location in the menu: **P3.1.2.15**

Use this parameter to set the motor stator phase-resistance value (in star-connection equivalent circuit). The resistance is essential for an accurate motor control.

The value is given as a percentage of voltage drop at nominal current.

NOTICE

The identification run sets a value for this parameter automatically. We recommend the identification run, if it is possible. Make the identification run with the parameter P3.1.2.4.

11.3.2.11 (ID 673) Stator Inductance Voltage Drop

Location in the menu: **P3.1.2.16**

Use this parameter to set the motor stator phase-inductance value (in star-connection equivalent circuit). The value is the transient inductance (for induction motors) or the line-to-neutral inductance (for permanent magnet and reluctance motors).

The value is given as a percentage of voltage drop at nominal current and nominal frequency.

NOTICE

The identification run sets a value for this parameter automatically. We recommend the identification run, if it is possible. Make the identification run with the parameter P3.1.2.4.

11.3.2.12 (ID 674) Motor BEM Voltage

Location in the menu: **P3.1.2.17**

Use this parameter to set the back electromotive force (back EMF) voltage of a permanent magnet motor. The value of this parameter is the motor terminal voltage when the motor current is zero.

The value is given as line-to-line RMS voltage at nominal operating frequency. Identification at standstill uses this parameter to set U/f curve in open loop control mode. The back EMF value that is identified in Identification with rotation overwrites the existing parameter value.

11.3.3 Motor Limits

11.3.3.1 (ID 107) Motor Current Limit

Location in the menu: **P3.1.3.1**

Use this parameter to set the maximum motor current from the AC drive. The range of values for the parameter is different for each enclosure size of the drive.

When the current limit is active, the drive output frequency decreases.

NOTICE

The Motor Current Limit is not an overcurrent trip limit.

11.3.3.2 (ID 1287) Motor Torque Limit

Location in the menu: **P3.1.3.2**

Use this parameter to set the maximum torque limit of the motoring side.

The range of values for the parameter is different for each enclosure size of the drive.

11.3.4 Open Loop Parameters

11.3.4.1 (ID 108) U/F Ratio

Location in the menu: **P3.1.4.1**

Use this parameter to set the type of the U/f curve between zero frequency and the field weakening point.

Selection number	Selection name	Description
0	Linear	The voltage of the motor changes linearly as a function of the output frequency. The voltage changes from the value of P3.1.4.6 (Zero Frequency Voltage) to the value of P3.1.4.3 (Voltage at Field Weakening Point) at a frequency set in P3.1.4.2 (Field Weakening Point Frequency). Use this default setting if a different setting is not necessary.
1	Squared	The voltage of the motor changes from the value of P3.1.4.6 (Zero Frequency Voltage) to the value of P3.1.4.2 (Field Weakening Point Frequency) at a squared curve. The motor operates undermagnetized below the field weakening point and produces less torque. Use the squared U/f ratio in applications where the torque demand is in relation to the square of the speed, for example in centrifugal fans and pumps.
2	Programmable	It is possible to program the U/f curve with 3 different points: the zero frequency voltage (P1), the mid-point voltage/ frequency (P2), and the field weakening point (P3). Use the programmable U/f curve at low frequencies if it is necessary to have more torque. Use the identification run (P3.1.2.4) to find the optimal settings automatically.

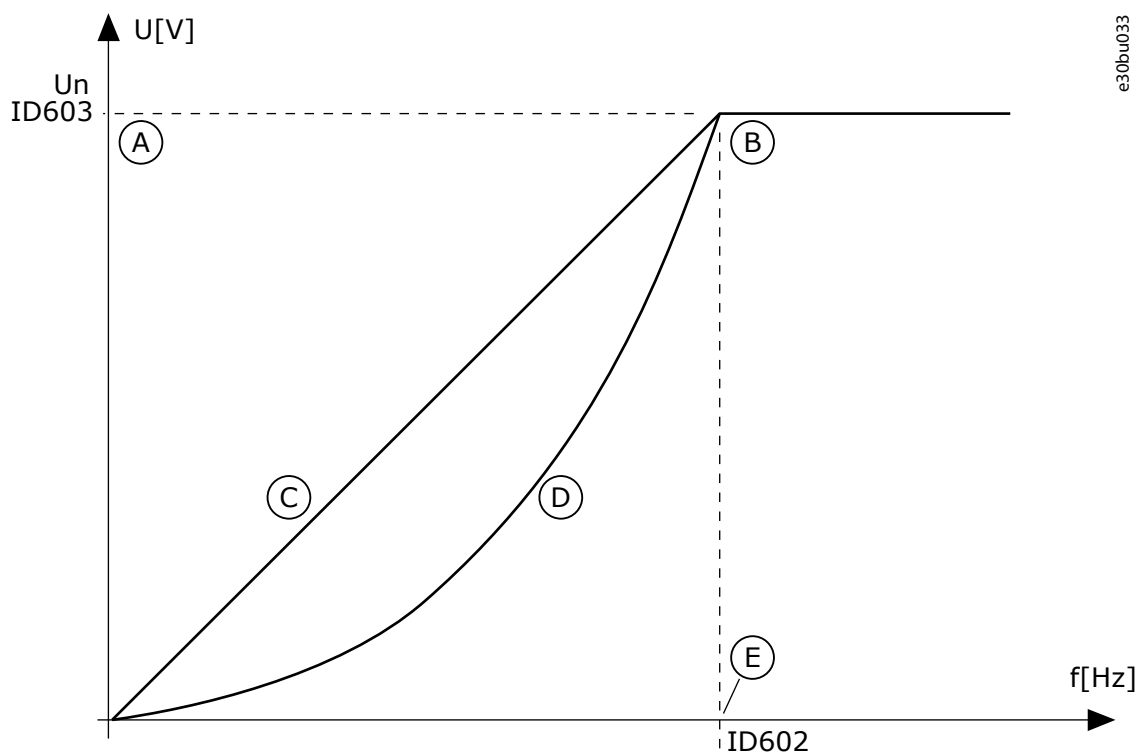


Illustration 37: Linear and Squared Change of the Motor Voltage

A	Default: Nominal voltage of the motor	D	Squared
B	Field weakening point	E	Default: Nominal frequency of the motor
C	Linear		

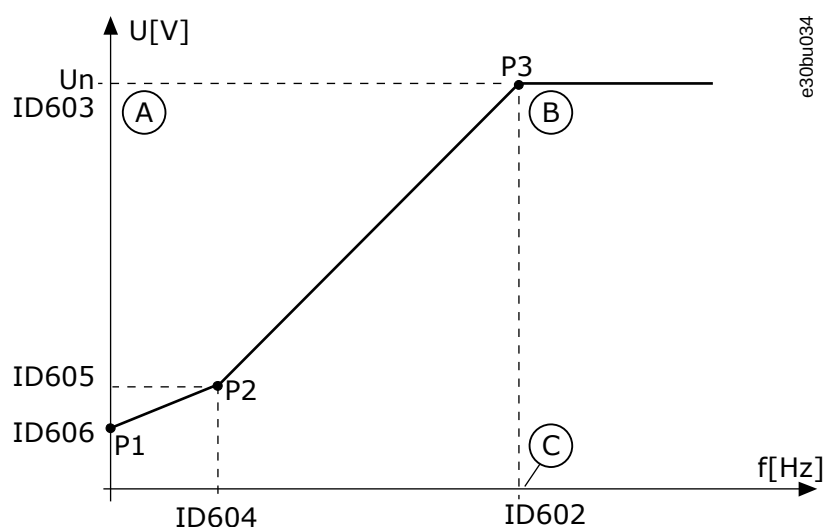


Illustration 38: The Programmable U/f Curve

A	Default: Nominal voltage of the motor	C	Default: Nominal frequency of the motor
B	Field weakening point		

When the parameter Motor Type has the value PM motor (Permanent Magnet Motor), this parameter is automatically set to the value Linear.

When the parameter Motor Type has the value Induction Motor, and when this parameter is changed, these parameters are set to their default values:

- P3.1.4.2 Field Weakening Point Frequency
- P3.1.4.3 Voltage at Field Weakening Point
- P3.1.4.4 U/f Midpoint Frequency
- P3.1.4.5 U/f Midpoint Voltage
- P3.1.4.6 Zero Frequency Voltage

11.3.4.2 (ID 602) Field Weakening Point Frequency

Location in the menu: **P3.1.4.2**

Use this parameter to set the output frequency at which the output voltage reaches the field weakening point voltage.

11.3.4.3 (ID 603) Voltage at Field Weakening Point

Location in the menu: **P3.1.4.3**

Use this parameter to set the voltage at the field weakening point as a percentage of the motor nominal voltage.

Above the frequency at the field weakening point, the output voltage stays at the set maximum value. Below the frequency at the field weakening point, the U/f curve parameters control the output voltage. See the U/f parameters P3.1.4.1, P3.1.4.4 and P3.1.4.5.

When setting the parameters P3.1.1.1 (Motor nominal voltage) and P3.1.1.2 (Motor nominal frequency), the parameters P3.1.4.2 and P3.1.4.3 automatically receive related values. To have different values for P3.1.4.2 and P3.1.4.3, change these parameters only after setting the parameters P3.1.1.1 and P3.1.1.2.

11.3.4.4 (ID 604) U/f Midpoint Frequency

Location in the menu: **P3.1.4.4**

Use this parameter to set the middle point frequency of the U/f curve.

NOTICE

This parameter gives the middle point frequency of the curve if the value of P3.1.4.1 is programmable.

11.3.4.5 (ID 605) U/f Midpoint Voltage

Location in the menu: **P3.1.4.5**

Use this parameter to set the middle point voltage of the U/f curve.

N O T I C E

This parameter gives the middle point voltage of the curve if the value of P3.1.4.1 is programmable.

11.3.4.6 (ID 606) Zero Frequency Voltage

Location in the menu: **P3.1.4.6**

Use this parameter to set the zero frequency voltage of the U/f curve. The default value for the parameter is different for each unit size.

11.3.4.7 (ID 1590) Flying Start Options

Location in the menu: **P3.1.4.7**

Use this parameter to set the flying start options. The parameter Flying Start Options has a checkbox selection of values.

The bits can receive these values.

- Search the shaft frequency only from the same direction as the frequency reference
- Disable the AC scanning
- Use the frequency reference for an initial guess
- Disable the DC pulses
- Flux build with current control

The bit B0 controls the search direction. With the bit set to 0, the shaft frequency is searched in 2 directions, the positive and the negative. With the bit set to 1, the shaft frequency is searched only in the frequency reference direction. It prevents the shaft movements for the other direction.

The bit B1 controls the AC scanning that premagnetizes the motor. In the AC scanning, the system sweeps the frequency from the maximum towards zero frequency. The AC scanning stops when an adaptation to the shaft frequency occurs. To disable the AC scanning, set the bit B1 to 1. If the value of Motor Type is permanent magnet motor, the AC scanning is disabled automatically.

Use the bit B5 to disable the DC pulses. The primary function of the DC pulses is to premagnetize the motor and examine the rotation of the motor. If the DC pulses and the AC scanning are enabled, the slip frequency tells which procedure is applied. If the slip frequency is less than 2 Hz, or the motor type is PM motor, the DC pulses are disabled automatically.

The bit B7 controls the rotation direction of the injected high frequency signal, which is used in the flying start of synchronous reluctance machines. Signal injection is used to detect the frequency of the rotor. If the rotor is in a blind angle when the signal is injected, the rotor frequency is undetectable. Reversing the rotation direction of the injection signal solves this problem.

11.3.4.8 (ID 1610) Flying Start Scan Current

Location in the menu: **P3.1.4.8**

Use this parameter to set the flying start scan current as a percentage of the motor nominal current.

11.3.4.9 (ID 109) Start Boost

Location in the menu: **P3.1.4.9**

Use this parameter with a process that has a high starting torque because of friction.

Use the start boost only when starting the drive. The start boost is deactivated after 10 s or when the output frequency of the drive is more than half of the frequency of the field weakening point.

The voltage to the motor changes in relation to the necessary torque. It makes the motor give more torque at the start and when the motor operates at low frequencies.

The start boost affects a linear U/f curve. To get the best result, do the identification run and activate the programmable U/f curve.

11.3.5 I/f Start Function

With a PM motor, use the I/f Start function to start the motor with constant current control. A high power motor gives the best effect. With a high power motor, the resistance is low and it is not easy to change the U/f curve.

The I/f Start function can also give a sufficient torque for the motor at start up.

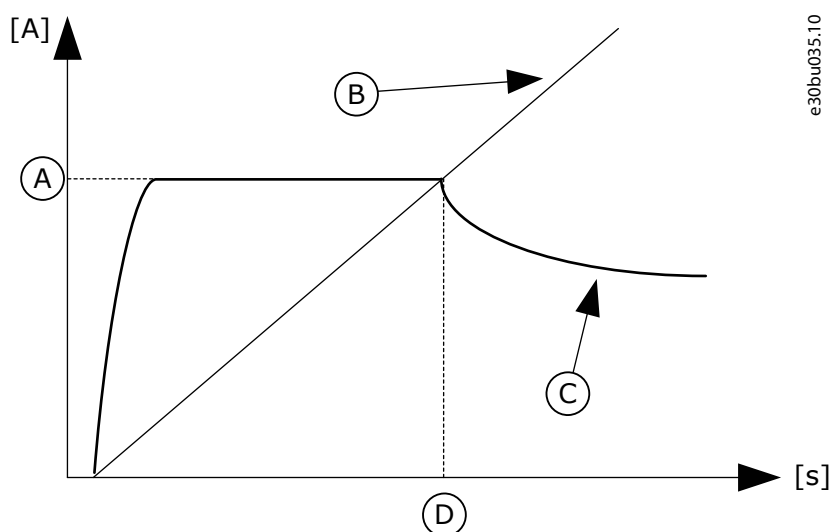


Illustration 39: The I/f Start Parameters

A	I/f Start Current	C	Motor Current
B	Output Frequency	D	I/f Start Frequency

11.3.5.1 (ID 534) I/f Start

Location in the menu: **P3.1.4.12.1**

Use this parameter to enable the I/f start function.

When the I/f start function is activated, the drive starts to operate in the current control mode. A constant current is led to the motor until the output frequency increases above the level that is set in P3.1.4.12.2. When the output frequency increases above I/f Start Frequency level, the operation mode changes back to the normal U/f control mode.

11.3.5.2 (ID 535) I/f Start Frequency

Location in the menu: **P3.1.4.12.2**

Use this parameter to set the output frequency limit below which the set I/f start current is fed to motor.

When the output frequency of the drive is below the limit of this parameter, I/f start function activates. When the output frequency is more than the limit, the drive operation mode changes back to the normal U/f control mode.

11.3.5.3 (ID 536) I/f Start Current

Location in the menu: **P3.1.4.12.3**

Use this parameter to set the current that is used when the I/f start function is enabled.

11.3.6 Torque Stabilator Function

11.3.6.1 (ID 1412) Torque Stabilator Gain

Location in the menu: **P3.1.4.13.1**

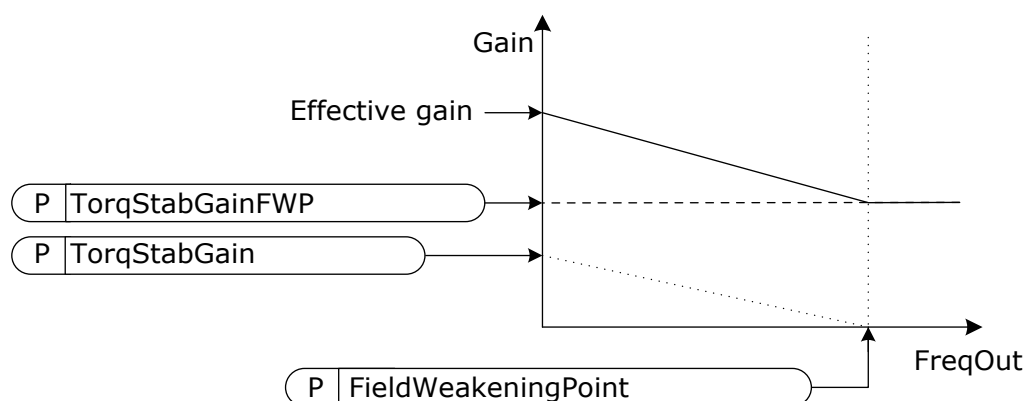
Use this parameter to set the gain of the torque stabilator in an open loop control operation.

11.3.6.2 (ID 1414) Torque Stabilator Gain at Field Weakening Point

Location in the menu: **P3.1.4.13.2**

Use this parameter to set the gain of the torque stabilator at field weakening point in an open loop control operation. The torque stabilizer stabilizes the possible oscillations in the estimated torque.

Two gains are used. TorqStabGainFWP is a constant gain at all the output frequencies. TorqStabGain changes linearly between the zero frequency and the field weakening point frequency. The full gain is at 0 Hz and the gain is zero at the field weakening point. The figure shows the gains as a function of output frequency.



e30bu036

Illustration 40: The Torque Stabilizer Gain

11.3.6.3 (ID 1413) Torque Stabilizer Damping Time Constant

Location in the menu: **P3.1.4.13.3**

Use this parameter to set the damping time constant of the torque stabilizer.

11.3.6.4 (ID 1735) Torque Stabilizer Damping Time Constant for PM Motors

Location in the menu: **P3.1.4.13.4**

Use this parameter to set the damping time constant of the torque stabilizer for permanent magnet or reluctance motors.

11.3.6.5 (ID 15650) Torque Stabilizer Filter Time Constant

Location in the menu: **P3.1.4.13.5**

Use this parameter to set the lowpass filtering time constant of the torque stabilizer.

11.3.6.6 (ID 15651) Torque Stabilizer Filter Time Constant for PM Motors

Location in the menu: **P3.1.4.13.6**

Use this parameter to set the lowpass filtering time constant of the torque stabilizer for permanent magnet or reluctance motors.

11.4 Start/Stop Setup

11.4.1 Start and Stop Commands in Different Control Places

Start and stop commands must be given differently in each control place.

Remote Control Place (I/O A)

Use the parameters P3.5.1.1 (Control signal 1 A), P3.5.1.2 (Control signal 2 A) and P3.5.1.3 (Control signal 3 A) to select digital inputs. These digital inputs control the start, stop, and reverse commands. Then select a logic for these inputs with P3.2.6 I/O A Logic.

Remote Control Place (I/O B)

Use the parameters P3.5.1.4 (Control signal 1 B), P3.5.1.5 (Control signal 2 B) and P3.5.1.6 (Control signal 3 B) to select digital inputs. These digital inputs control the start, stop, and reverse commands. Then select a logic for these inputs with P3.2.7 I/O B Logic.

Local Control Place (Keypad)

The start and stop commands come from the keypad buttons. The direction of the rotation is set with parameter P3.3.1.9 Keypad direction.

Remote Control Place (Fieldbus)

Start, stop, and reverse commands come from the fieldbus.

11.4.2 (ID 172) Remote Control Place

Location in the menu: **P3.2.1**

Use this parameter to select the remote control place (start/stop). Use this parameter to change back to remote control from VACON® Live, for example if the control panel is broken.

11.4.3 (ID 211) Local/Remote

Location in the menu: **P3.2.2**

Use this parameter to switch between the local and remote control places. Local control place is always keypad control. The remote control place can be I/O or Fieldbus, depending on the 'Remote Control Place' parameter value.

11.4.4 (ID 114) Keypad Stop Button

Location in the menu: **P3.2.3**

Use this parameter to enable the keypad stop button. When this function is enabled, a press of keypad stop button always stops the drive (regardless of the control place). When this function is disabled, a press of keypad stop button stops the drive in local control only.

Selection number	Selection name	Description
0	Yes	The keypad stop button is always enabled.
1	No	Limited function of the keypad stop button.

11.4.5 (ID 505) Start Function

Location in the menu: **P3.2.4**

Use this parameter to select the type of the start function.

Selection number	Selection name	Description
0	Ramping	The drive accelerates from 0 frequency to frequency reference.
1	Flying start	The drive detects the actual speed of the motor and accelerates from that speed to frequency reference.

11.4.6 (ID 506) Stop Function

Location in the menu: **P3.2.5**

Use this parameter to select the type of the stop function.

Selection number	Selection name	Description
0	Coasting	The motor stops on its inertia. When the stop command is given, the control by the drive stops and the current from the drive goes to 0.
1	Ramp	After the stop command, the speed of the motor is decreased to zero speed according to the deceleration parameters.

NOTICE

Ramp stop cannot be guaranteed in all situations. If ramp stop is selected and the net voltage changes over 20%, the voltage estimation fails. In such case, ramp stop is not possible.

11.4.7 (ID 300) I/O A Start/Stop Logic

Location in the menu: **P3.2.6**

Use this parameter to control the start and stop of the drive with the digital signals. The selections can include the word 'edge' to help preventing an accidental start.

An accidental start can occur, for example, in these conditions:

- When the power is connected.
- When the power is connected again after a power cut.
- After resetting a fault.
- After Run Enable stops the drive.
- When changing the control place to I/O control.

Before starting the motor, the Start/Stop contact must be opened.

In all the examples of the next pages, the stop mode is coasting. CS = Control signal.

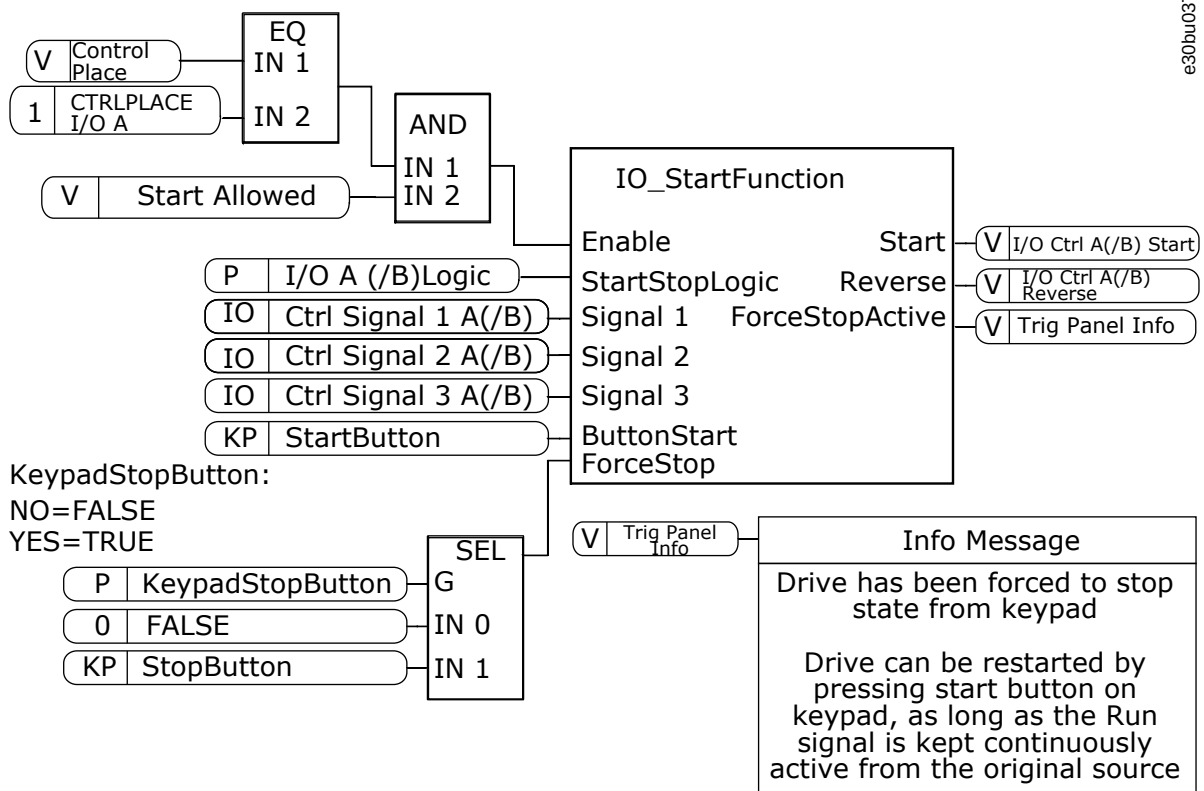


Illustration 41: The Block Diagram of the I/O A Start/Stop Logic

Selection = 0

Selection number	Selection name	Description
0	CS1 = Forward CS2 = Backward	The functions activate when the contacts are closed.

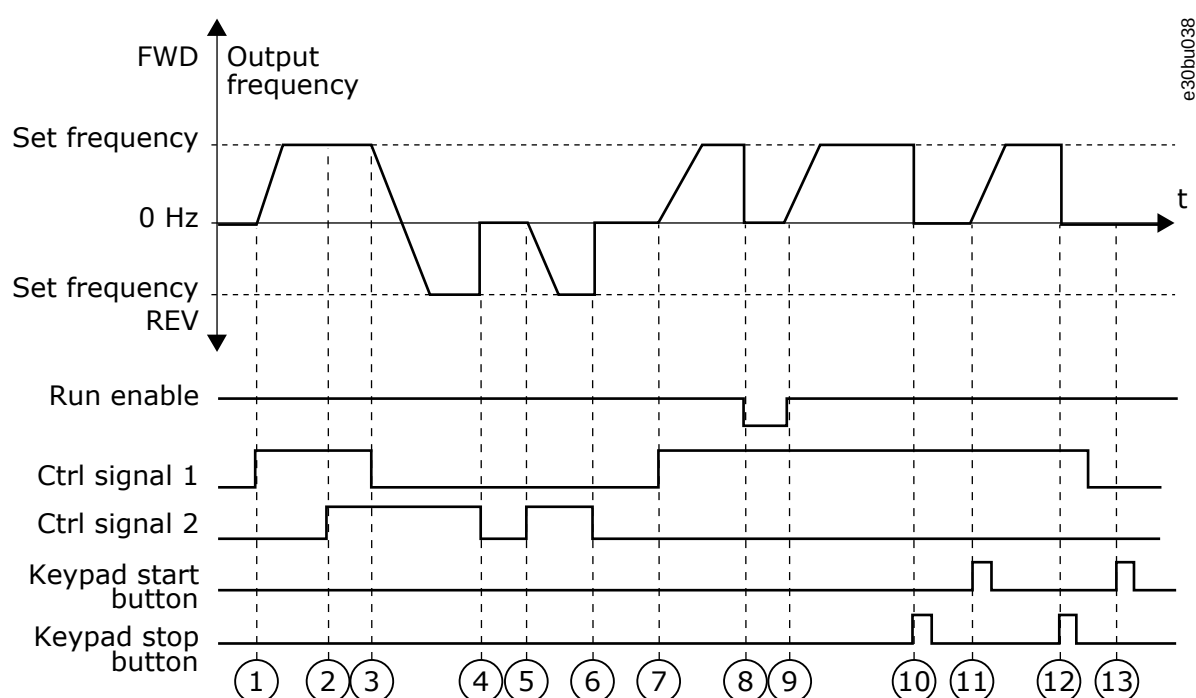


Illustration 42: I/O A Start/Stop Logic = 0

1	Control signal (CS) 1 activates and causes the output frequency to increase. The motor operates forward.	8	The Run enable signal is set to OPEN, which causes the frequency to go to 0. Configure the Run enable signal with parameter P3.5.1.15.
2	CS2 activates, but it does not affect the output frequency, because the direction that is set first has the highest priority.	9	The Run enable signal is set to CLOSED, which causes the frequency to increase to the set frequency, because CS1 is still active.
3	CS1 becomes inactive and causes the direction to start to change (FWD to REV), because CS2 is still active.	10	The STOP button on the keypad is pushed, and the frequency that is fed to the motor goes to 0. (This signal only works if the value of P3.2.3 Keypad Stop Button is Yes.)
4	CS2 becomes inactive and the frequency that is fed to the motor goes to 0.	11	The drive starts because the START button on the keypad was pushed.
5	CS2 activates again and causes the motor to accelerate (REV) to the set frequency.	12	The STOP button on the keypad is pushed again to stop the drive.
6	CS2 becomes inactive and the frequency fed to the motor drops to 0.	13	The attempt to start the drive with the START button is not successful, because CS1 is inactive.
7	CS1 activates and the motor accelerates (FWD) to the set frequency.		

Selection = 1

Selection number	Selection name	Description
1	CS1 = Forward (edge) CS2 = Inverted stop CS3 = Backward (edge)	For a 3-wire control (pulse control)

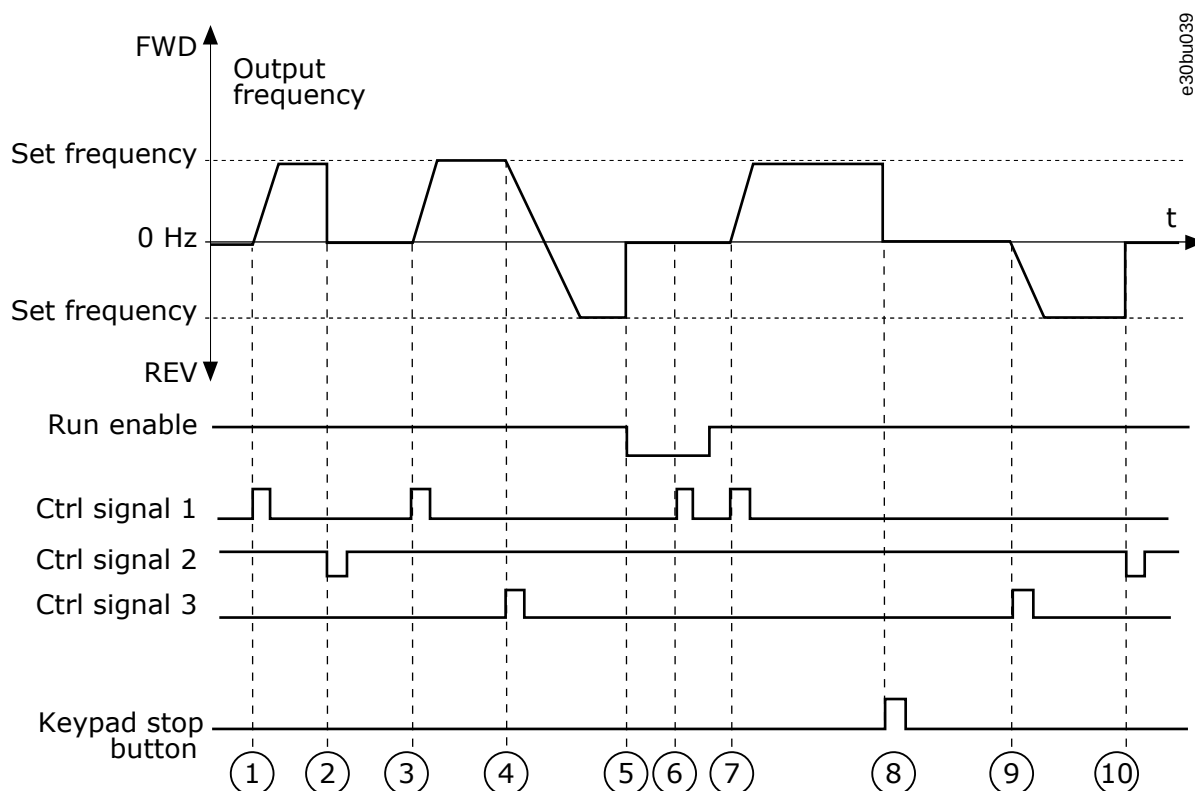


Illustration 43: I/O A Start/Stop Logic = 1

1	Control signal (CS) 1 activates and causes the output frequency to increase. The motor operates forward.	6	The start attempt with CS1 is not successful, because the Run enable signal is still OPEN.
2	CS2 becomes inactive and causes the frequency to go to 0.	7	CS1 activates and the motor accelerates (FWD) to the set frequency, because the Run enable signal was set to CLOSED.
3	CS1 activates and causes the output frequency to increase again. The motor operates forward.	8	The STOP button on the keypad is pushed, and the frequency that is fed to the motor goes to 0. (This signal only works if the value of P3.2.3 Keypad Stop Button is Yes.)
4	CS3 activates and causes the direction to start to change (FWD to REV).	9	CS3 activates and causes the motor to start and to operate in the reverse direction.
5	The Run enable signal is set to OPEN, which causes the frequency to go to 0. Configure the Run enable signal with parameter 3.5.1.15.	10	CS2 becomes inactive and causes the frequency to go to 0.

Selection = 2

Selection number	Selection name	Description
2	CS1 = Forward (edge) CS2 = Backward (edge)	Use this function to prevent an accidental start. Before starting the motor again, the start/stop contact must be opened.

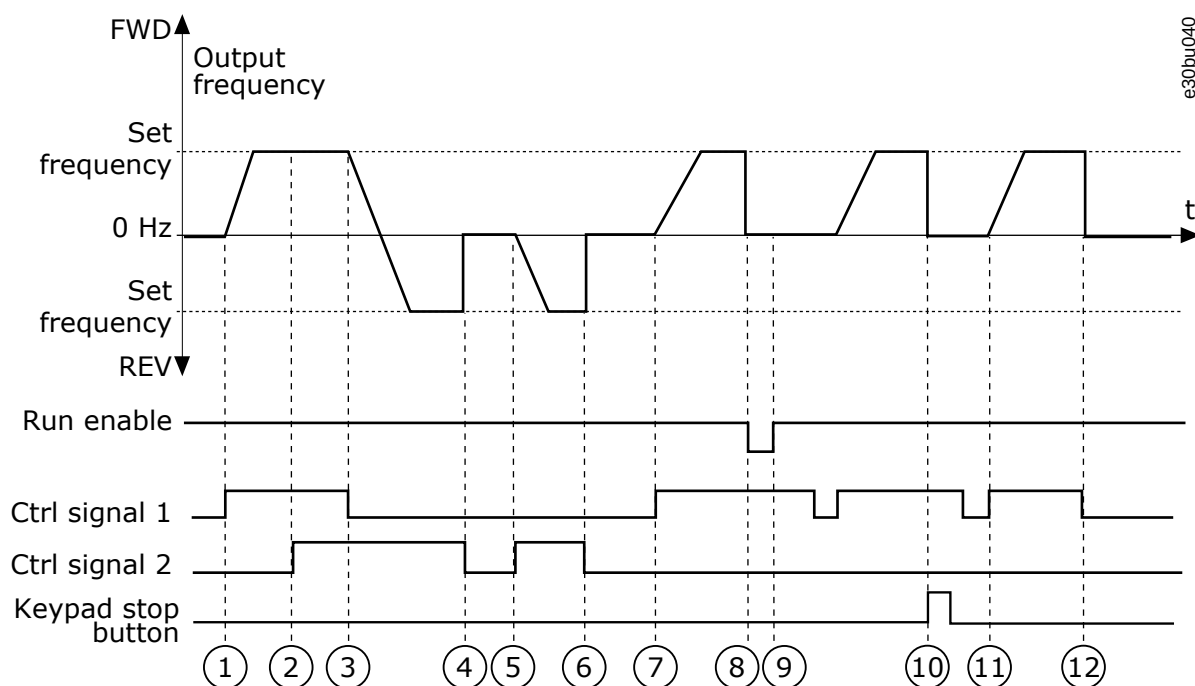


Illustration 44: I/O A Start/Stop Logic = 2

1	Control signal (CS) 1 activates and causes the output frequency to increase. The motor operates forward.	7	CS1 activates and the motor accelerates (FWD) to the set frequency.
2	CS2 activates, but it does not affect the output frequency, because the direction that is set first has the highest priority.	8	The Run enable signal is set to OPEN, which causes the frequency to go to 0. Configure the Run enable signal with parameter P3.5.1.15.
3	CS1 becomes inactive and causes the direction to start to change (FWD to REV), because CS2 is still active.	9	The Run enable signal is set to CLOSED, which does not affect, because a rising edge is necessary for the start, even if CS1 is active.
4	CS2 becomes inactive and the frequency that is fed to the motor goes to 0.	10	The STOP button on the keypad is pushed and the frequency that is fed to the motor goes to 0. (This signal only works if the value of P3.2.3 Keypad Stop Button is Yes.)
5	CS2 activates again and causes the motor to accelerate (REV) to the set frequency.	11	CS1 is opened and closed again, which causes the motor to start.
6	CS2 becomes inactive and the frequency that is fed to the motor goes to 0.	12	CS1 becomes inactive and the frequency that is fed to the motor goes to 0.

Selection = 3

Selection number	Selection name	Description
3	CS1 = Start CS2 = Reverse	

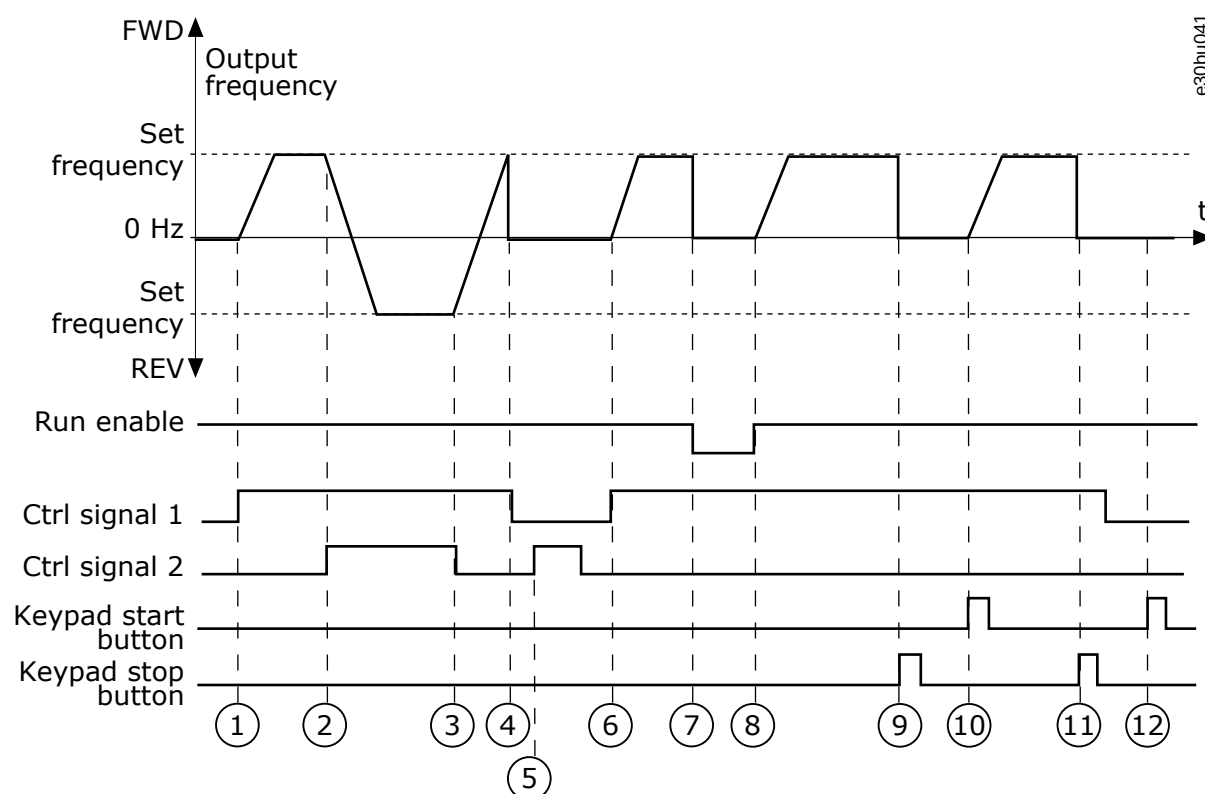


Illustration 45: I/O A Start/Stop Logic = 3

1	Control signal (CS) 1 activates and causes the output frequency to increase. The motor operates forward.	7	The Run enable signal is set to OPEN, which causes the frequency to go to 0. Configure the Run enable signal with parameter P3.5.1.15.
2	CS2 activates and causes the direction to start to change (FWD to REV).	8	The Run enable signal is set to CLOSED, which causes the frequency to increase to the set frequency, because CS1 is still active.
3	CS2 becomes inactive, which causes the direction to start to change (REV to FWD), because CS1 is still active.	9	The STOP button on the keypad is pushed and the frequency that is fed to the motor goes to 0. (This signal only works if the value of P3.2.3 Keypad Stop Button is Yes.)
4	CS1 becomes inactive and the frequency goes to 0.	10	The drive starts because the START button on the keypad was pushed.
5	CS2 activates, but the motor does not start because CS1 is inactive.	11	The drive is stopped again with the STOP button on the keypad.
6	CS1 activates and causes the output frequency to increase again. The motor operates forward because CS2 is inactive.	12	The attempt to start the drive with the START button is not successful, because CS1 is inactive.

Selection = 4

Selection number	Selection name	Description
4	CS1 = Start (edge) CS2 = Reverse	Use this function to prevent an accidental start. Before starting the motor again, the start/stop contact must be opened.

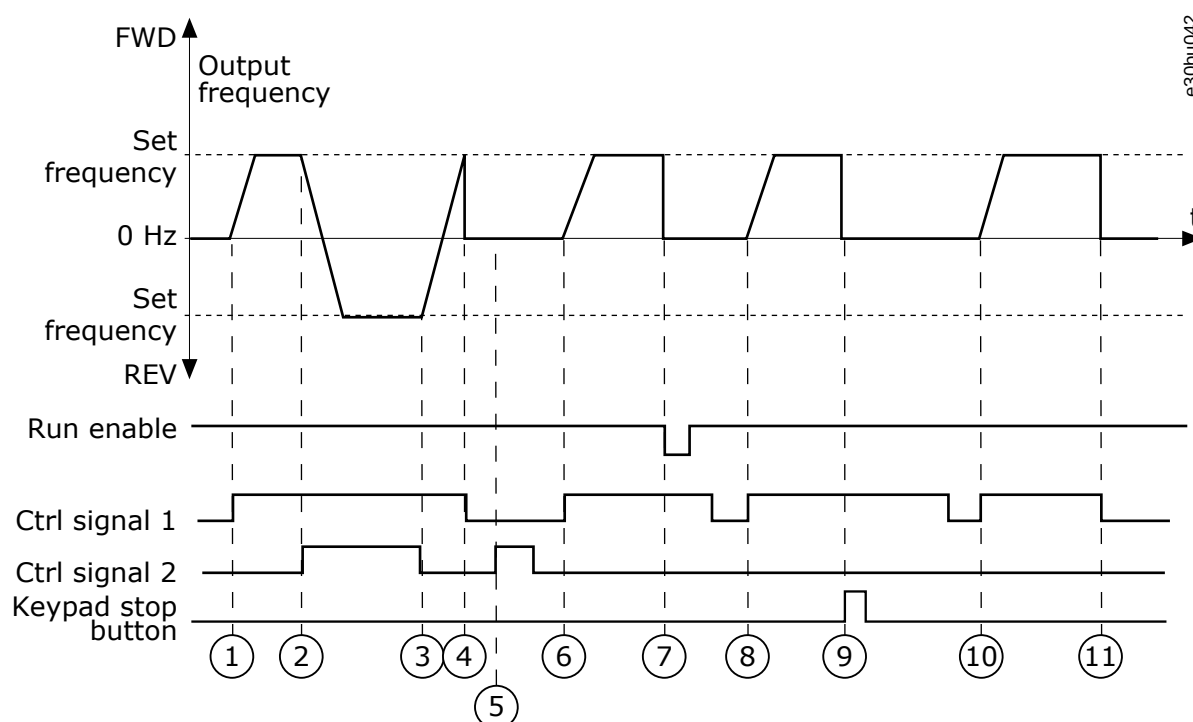


Illustration 46: I/O A Start/Stop Logic = 4

1	Control signal (CS) 1 activates and causes the output frequency to increase. The motor operates forward because CS2 is inactive.	7	The Run enable signal is set to OPEN, which causes the frequency to go to 0. Configure the Run enable signal with parameter P3.5.1.15.
2	CS2 activates, which causes the direction to start to change (FWD to REV).	8	Before the drive can start, CS1 must be opened and closed again.
3	CS2 becomes inactive, which causes the direction to start to change (REV to FWD), because CS1 is still active.	9	The STOP button on the keypad is pushed and the frequency that is fed to the motor goes to 0. (This signal only works if the value of P3.2.3 Keypad Stop Button is Yes.)
4	CS1 becomes inactive and the frequency goes to 0.	10	Before the drive can start, CS1 must be opened and closed CS1 again.
5	CS2 activates, but the motor does not start because CS1 is inactive.	11	CS1 becomes inactive and the frequency goes to 0.
6	CS1 activates and causes the output frequency to increase again. The motor operates forward, because CS2 is inactive.		

11.4.8 (ID 363) I/O B Start/Stop Logic

Location in the menu: **P3.2.7**

Use this parameter to control the start and stop of the drive with the digital signals. The selections can include the word 'edge' to help to prevent an accidental start. See P3.2.6 for more information.

11.4.9 (ID 889) Fieldbus Start Logic

Location in the menu: **P3.2.8**

Use this parameter to set the fieldbus start logic. The selections can include the word 'edge' to help to prevent an accidental start.

Selection number	Selection name	Description
0	A rising edge is necessary	
1	State	

11.4.10 (ID 524) Start Delay

Location in the menu: **P3.2.9**

Use this parameter to set the delay between the start command and the actual start of the drive.

11.4.11 (ID 181) Remote to Local Function

Location in the menu: **P3.2.10**

Use this parameter to set the selection of copy settings when going from Remote to Local (keypad) control.

Selection number	Selection name	Description
0	Keep Run	
1	Keep Run & Reference	
2	Stop	

11.4.12 (ID 15555) Restart Delay

Location in the menu: **P3.2.11**

Use this parameter to set the time delay during which the drive cannot be restarted after the drive has been stopped. The parameter is used in compressor applications.

Selection number	Selection name	Description
0	Restart delay not used	

11.5 References

11.5.1 Frequency Reference

It is possible to program the source of the frequency reference in all the control places, except the PC tool. When using PC, it always takes the frequency reference from the PC tool.

Remote Control Place (I/O A)

To set the source of the frequency reference for I/O A, use the parameter P3.3.1.5.

Remote Control Place (I/O B)

To set the source of the frequency reference for I/O B, use the parameter P3.3.1.6.

Local Control Place (Keypad)

If the default value keypad for the parameter P3.3.1.7 is used, the reference that was set for P3.3.1.8 Keypad Reference applies.

Remote Control Place (Fieldbus)

If the default value fieldbus for the parameter P3.3.1.10 is used, the frequency reference comes from fieldbus.

11.5.1.1 (ID 101) Minimum Frequency Reference

Location in the menu: **P3.3.1.1**

Use this parameter to set the minimum frequency reference.

11.5.1.2 (ID 102) Maximum Frequency Reference

Location in the menu: **P3.3.1.2**

Use this parameter to set the maximum frequency reference.

11.5.1.3 (ID 1285) Positive Frequency Reference Limit

Location in the menu: **P3.3.1.3**

Use this parameter to set the final frequency reference limit for the positive direction.

11.5.1.4 (ID 1286) Negative Frequency Reference Limit

Location in the menu: **P3.3.1.4**

Use this parameter to set the final frequency reference limit for the negative direction. Use this parameter for example to prevent the motor from running in the reverse direction.

11.5.1.5 (ID 117) I/O Control Reference A Selection

Location in the menu: **P3.3.1.5**

Use this parameter to select the reference source when the control place is I/O A. The application set with parameter 1.2 gives the default value.

11.5.1.6 (ID 131) I/O Control Reference B Selection

Location in the menu: **P3.3.1.6**

Use this parameter to select the reference source when the control place is I/O B. See P3.3.1.5 for more information. It is possible to force the I/O B control place to be active only with a digital input (P3.5.1.7).

11.5.1.7 (ID 121) Keypad Control Reference Selection

Location in the menu: **P3.3.1.7**

Use this parameter to select the reference source when the control place is keypad.

11.5.1.8 (ID 184) Keypad Reference

Location in the menu: **P3.3.1.8**

Use this parameter to adjust the frequency reference on the keypad.

11.5.1.9 (ID 123) Keypad Direction

Location in the menu: **P3.3.1.9**

Use this parameter to set the rotation direction of the motor when the control place is keypad.

11.5.2 Preset Frequencies

Use the Preset frequencies function in processes where more than 1 fixed frequency reference is necessary. There are 8 preset frequency references available. The selection of a preset frequency reference can be made with the digital input signals P3.3.3.10, P3.3.3.11, and P3.3.3.12.

11.5.2.1 (ID 182) Preset Frequency Mode

Location in the menu: **P3.3.3.1**

Use this parameter to set the logic of the digital input preset frequencies.

With this parameter, it is possible to set the logic which one of the preset frequencies is selected into use. There is a selection of 2 different logics. The number of preset speed digital inputs that are active defines the preset frequency.

Selection number	Selection name	Description
0	Binary coded	The mix of the inputs is binary coded. The different sets of active digital inputs determine the preset frequency.
1	Number (of inputs used)	The number of active inputs tells which preset frequency is used: 1, 2 or 3.

Value 0 selected

To set Preset Frequency 0 as reference, set the value 0 *Preset Frequency 0* for P3.3.1.5 (see [11.5.1.5 \(ID 117\) I/O Control Reference A Selection](#)).

To select a preset frequency 1–7, give digital inputs to P3.3.3.10 ([11.5.2.10 \(ID 419\) Preset Frequency Selection 0](#)), P3.3.3.11 ([11.5.2.11 \(ID 420\) Preset Frequency Selection 1](#)), and/or P3.3.3.12 ([11.5.2.12 \(ID 421\) Preset Frequency Selection 2](#)). The different sets of active digital inputs determine the preset frequency. For more data, see the following table. The values of the preset frequencies stay automatically between the minimum and maximum frequencies (P3.3.1.1 and P3.3.1.2).

Necessary step	Activated frequency
Select the value 0 for parameter P3.3.1.5.	Preset frequency 0

Table 128: The Selection of Preset Frequencies when Preset Frequency Mode is *Binary Coded*

Digital input signal Pre-set Freq Sel2 (P3.3.3.12)	Digital input signal Pre-set Freq Sel1 (P3.3.3.11)	Digital input signal Pre-set Freq Sel0 (P3.3.3.10)	Activated frequency reference
			Preset frequency 0 Only if Preset Freq 0 is set as frequency reference source with P3.3.3.1.5, P3.3.1.6, P3.3.1.7 or P3.3.1.10.
		Activated	Preset frequency 1
	Activated		Preset frequency 2
	Activated	Activated	Preset frequency 3
Activated			Preset frequency 4
Activated		Activated	Preset frequency 5
Activated	Activated		Preset frequency 6
Activated	Activated	Activated	Preset frequency 7

Value 1 selected

It is possible to use the Preset Frequencies 1–3 with different sets of active digital inputs. The number of active inputs tells which one is used.

Table 129: The Selection of Preset Frequencies when Preset Frequency Mode is *Number of Inputs*

Digital input signal Pre-set Freq Sel2 (P3.3.3.12)	Digital input signal Pre-set Freq Sel1 (P3.3.3.11)	Digital input signal Pre-set Freq Sel0 (P3.3.3.10)	Activated frequency reference
			Preset frequency 0 Only if Preset Freq 0 is set as frequency reference source with P3.3.3.1.5, P3.3.1.6, P3.3.1.7 or P3.3.1.10.
		Activated	Preset frequency 1
	Activated		Preset frequency 1
Activated			Preset frequency 1
	Activated	Activated	Preset frequency 2
Activated		Activated	Preset frequency 2
Activated	Activated		Preset frequency 2
Activated	Activated	Activated	Preset frequency 3

11.5.2.2 (ID 180) Preset Frequency 0

Location in the menu: **P3.3.3.2**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.3 (ID 105) Preset Frequency 1

Location in the menu: **P3.3.3.3**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.4 (ID 106) Preset Frequency 2

Location in the menu: **P3.3.3.4**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.5 (ID 126) Preset Frequency 3

Location in the menu: **P3.3.3.5**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.6 (ID 127) Preset Frequency 4

Location in the menu: **P3.3.3.6**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.7 (ID 128) Preset Frequency 5

Location in the menu: **P3.3.3.7**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.8 (ID 129) Preset Frequency 6

Location in the menu: **P3.3.3.8**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.9 (ID 130) Preset Frequency 7

Location in the menu: **P3.3.3.9**

Use this parameter to set the preset frequency reference when the preset frequencies function is used. Select the preset frequencies with the digital input signals.

See more information in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#).

11.5.2.10 (ID 419) Preset Frequency Selection 0

Location in the menu: **P3.3.3.10**

Use this parameter to select the digital input signal that is used as a selector for the preset frequencies. This parameter is a binary selector for Preset speeds (0–7). See parameters P3.3.3.2 to P3.3.3.9.

11.5.2.11 (ID 420) Preset Frequency Selection 1

Location in the menu: **P3.3.3.11**

Use this parameter to select the digital input signal that is used as a selector for the preset frequencies. This parameter is a binary selector for Preset speeds (0–7). See parameters P3.3.3.2 to P3.3.3.9.

11.5.2.12 (ID 421) Preset Frequency Selection 2

Location in the menu: **P3.3.3.12**

Use this parameter to select the digital input signal that is used as a selector for the preset frequencies. This parameter is a binary selector for Preset speeds (0–7). See parameters P3.3.3.2 to P3.3.3.9.

To apply Preset frequencies 1–7, connect a digital input to these functions with the instructions in [11.7.1.2 Programming of Digital Inputs](#). See more data in [11.5.2.1 \(ID 182\) Preset Frequency Mode](#), and also in [Table 36](#), and [Table 44](#).

11.5.3 Motor Potentiometer Parameters

The frequency reference of the Motor Potentiometer is available in all the control places. It is possible to change the motor potentiometer reference only when the drive is in the run state.

NOTICE

If the output frequency is set slower than the Motor Potentiometer Ramp Time, the normal acceleration and deceleration times give limits to it.

11.5.3.1 (ID 418) Motor Potentiometer UP

Location in the menu: **P3.3.4.1**

Use this parameter to increase the output frequency with a digital input signal. With a motor potentiometer, it is possible to increase and decrease the output frequency. When a digital input is connected to parameter Motor Potentiometer UP, and the digital input signal is active, the output frequency rises. The motor potentiometer reference INCREASES until the contact is opened.

11.5.3.2 (ID 417) Motor Potentiometer DOWN

Location in the menu: **P3.3.4.2**

Use this parameter to decrease the output frequency with a digital input signal. With a motor potentiometer, it is possible to increase and decrease the output frequency. When a digital input is connected to parameter Motor Potentiometer DOWN, and the digital input signal is active, the output frequency falls. The motor potentiometer reference DECREASES until the contact is opened.

3 different parameters affect how the output frequency rises or falls when Motor Potentiometer UP or DOWN is active. These parameters are Motor Potentiometer Ramp Time (P3.3.4.3), Acceleration Time (P3.4.1.2), and Deceleration Time (P3.4.1.3).

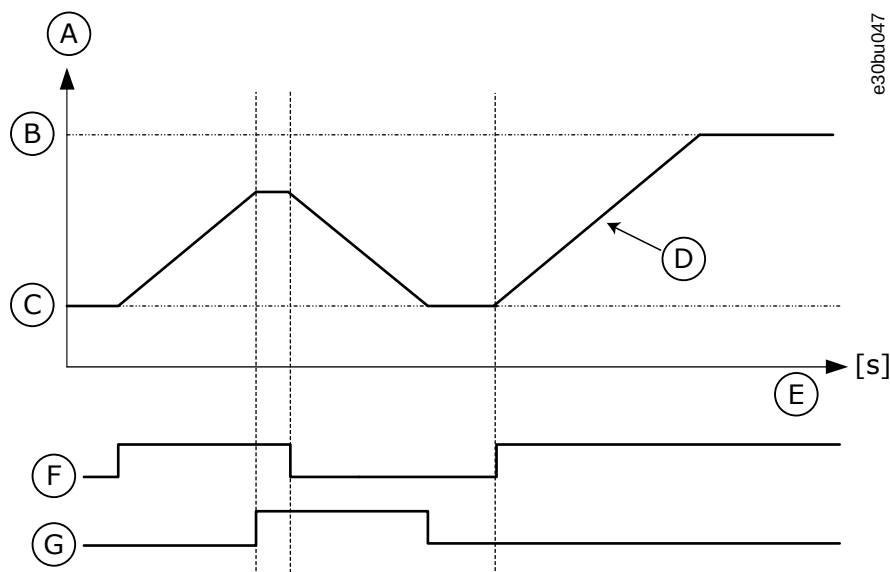


Illustration 47: The Motor Potentiometer Parameters

A	Frequency Reference	E	Time
B	Max Frequency	F	Motor potentiometer UP
C	Min Frequency	G	Motor potentiometer DOWN
D	Motor potentiometer ramp time		

11.5.3.3 (ID 331) Motor Potentiometer Ramp Time

Location in the menu: **P3.3.4.3**

Use this parameter to set the rate of change in the motor potentiometer reference when it is increased or decreased. The parameter value is entered as Hz/second.

11.5.3.4 (ID 367) Motor Potentiometer Reset

Location in the menu: **P3.3.4.4**

Use this parameter to set the logic for the resetting of the frequency reference of the motor potentiometer.

This parameter defines when the reference of the motor potentiometer is set to 0. There are 3 selections in the reset function: no reset, reset when the drive stops, or reset when the drive is powered down.

Selection number	Selection name	Description
0	No reset	The last motor potentiometer frequency reference is kept through the stop state and kept in memory when a power-down occurs.
1	Stop state	The motor potentiometer frequency reference is set to 0 when the drive goes to the stop state, or when the drive is powered down.
2	Powered down	The motor potentiometer frequency reference is set to 0 only when a power-down occurs.

11.5.4 Flushing Parameters

Use the Flushing function to override the normal control momentarily. Use the function to flush the pipeline or operate the pump manually at the preset constant speed, for example.

The Flushing function starts the drive at a selected reference without a start command no matter the control place.

11.5.4.1 (ID 530) Flushing Reference Activation

Location in the menu: **P3.3.6.1**

Use this parameter to select the digital input signal that activates the Flushing function. The flushing frequency reference is bidirectional and a reverse command does not affect the direction of the flushing reference.

N O T I C E

When the digital input is activated, the drive starts.

11.5.4.2 (ID 1239) Flushing Reference

Location in the menu: **P3.3.6.2**

Use this parameter to set the frequency reference of the drive when the Flushing function is used.

The reference is bidirectional and a reverse command does not affect the direction of the flushing reference. The reference for the forward direction is specified as a positive value and the reverse direction is specified as a negative value.

11.6 Ramps and Brakes Setup

11.6.1 Ramp 1

11.6.1.1 (ID 500) Ramp 1 Shape

Location in the menu: **P3.4.1.1**

Use this parameter to make the start and the end of the acceleration and deceleration ramps smoother.

With the parameters Ramp 1 Shape and Ramp 2 Shape, it is possible to make smoother the start and the end of the acceleration and deceleration ramps. The value is set to 0.0% gives a linear ramp shape. The acceleration and deceleration act immediately to the changes in the reference signal.

The value set between 1.0% and 100.0% gives an S-shaped acceleration or deceleration ramp. Use this function to reduce mechanical erosion of the parts and current spikes when the reference changes. The acceleration time can be modified with parameters P3.4.1.2 (Acceleration Time 1) and P3.4.1.3 (Deceleration Time 1).

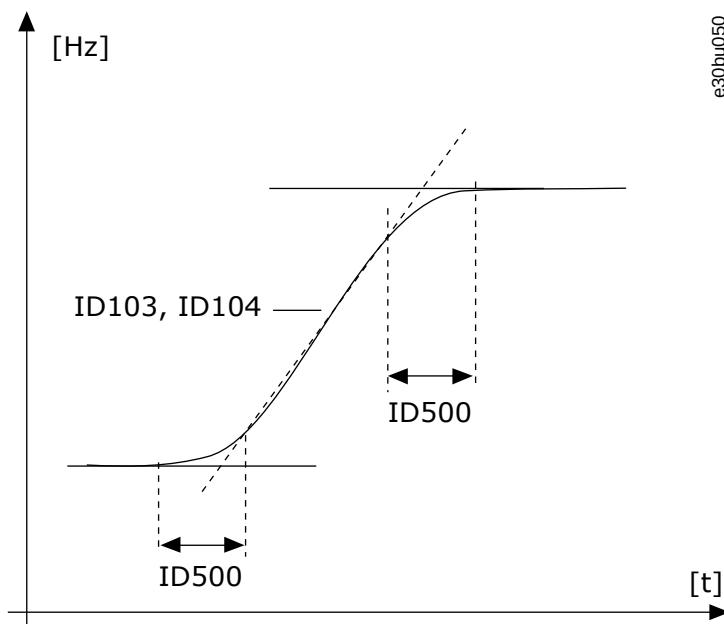


Illustration 48: The Acceleration/Deceleration Curve (S-shaped)

11.6.1.2 (ID 103) Acceleration Time 1

Location in the menu: **P3.4.1.2**

Use this parameter to set the time that is necessary for the output frequency to increase from zero frequency to maximum frequency.

11.6.1.3 (ID 104) Deceleration Time 1

Location in the menu: **P3.4.1.3**

Use this parameter to set the time that is necessary for the output frequency to decrease from maximum frequency to zero frequency.

11.6.2 Ramp 2

11.6.2.1 (ID 501) Ramp 2 Shape

Location in the menu: **P3.4.2.1**

Use this parameter to make the start and the end of the acceleration and deceleration ramps smoother.

With the parameters Ramp 1 Shape and Ramp 2 Shape, it is possible to make smoother the start and the end of the acceleration and deceleration ramps. The value set to 0.0% gives a linear ramp shape. The acceleration and deceleration act immediately to the changes in the reference signal.

The value set between 1.0% and 100.0% gives an S-shaped acceleration or deceleration ramp. Use this function to reduce mechanical erosion of the parts and current spikes when the reference changes. The acceleration time can be modified with parameters P3.4.2.2 (Acceleration Time 2) and P3.4.2.3 (Deceleration Time 2).

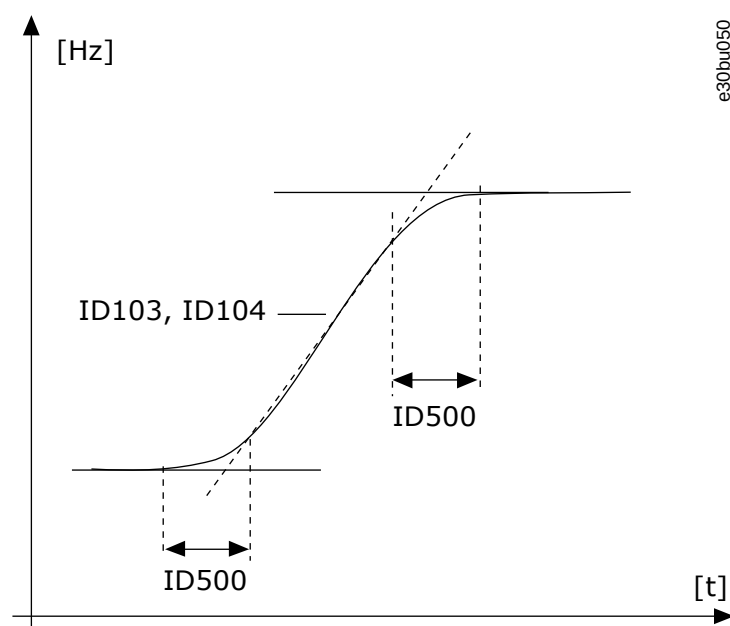


Illustration 49: The Acceleration/Deceleration Curve (S-shaped)

11.6.2.2 (ID 502) Acceleration Time 2

Location in the menu: **P3.4.2.2**

Use this parameter to set the time that is necessary for the output frequency to increase from zero frequency to maximum frequency.

11.6.2.3 (ID 503) Deceleration Time 2

Location in the menu: **P3.4.2.3**

Use this parameter to set the time that is necessary for the output frequency to decrease from maximum frequency to zero frequency.

11.6.2.4 (ID 408) Ramp 2 Selection

Location in the menu: **P3.4.2.4**

Use this parameter to select either ramp 1 or ramp 2.

Selection number	Selection name	Description
0	OPEN	Ramp 1 Shape, Acceleration Time 1, and Deceleration Time 1
1	CLOSED	Ramp 2 Shape, Acceleration Time 2, and Deceleration Time 2

11.6.2.5 (ID 533) Ramp 2 Threshold Frequency

Location in the menu: **P3.4.2.5**

Use this parameter to set the output frequency limit above which the Ramp 2 is used.

Use the function, for example, in applications for deep well pumps, where faster ramp times are necessary when the pump starts or stops (operates below the minimum frequency).

Second ramp times are activated when the output frequency of the drive goes above the limit specified by this parameter. To disable the function, set the value of the parameter to 0.

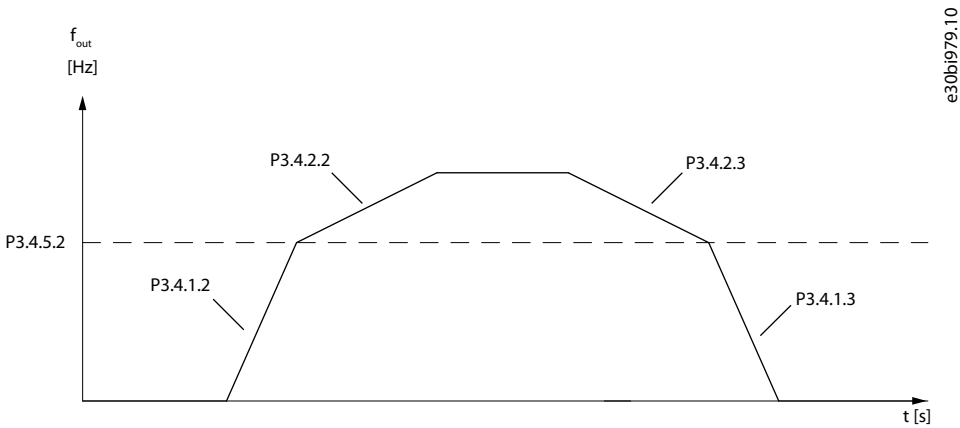


Illustration 50: Ramp 2 Activation when Output Frequency above the Threshold Level

P3.4.5.2 Ramp threshold freq	P3.4.1.3 Dec. time 1
P3.4.1.2 Acc. time 1	P3.4.2.3 Dec. time 2
P3.4.2.2 Acc. time 2	

11.6.3 Start Magnetizing

11.6.3.1 (ID 517) Start Magnetizing Current

Location in the menu: **P3.4.3.1**

Use this parameter to set the DC current that is fed into the motor at the start. If the value of this parameter is set to 0, the Start Magnetizing function is disabled.

11.6.3.2 (ID 516) Start Magnetizing Time

Location in the menu: **P3.4.3.2**

Use this parameter to set the time during which the DC current is fed to the motor before the acceleration starts.

11.6.4 DC Brake

11.6.4.1 (ID 507) DC Brake Current

Location in the menu: **P3.4.4.1**

Use this parameter to set the current that is fed into the motor during DC braking. If the value of this parameter is set to 0, the DC brake function is disabled.

11.6.4.2 (ID 508) DC Braking Time at Stop

Location in the menu: **P3.4.4.2**

Use this parameter to set the braking is ON or OFF and to give the braking time when the motor stops. If the value of this parameter is set to 0, the DC brake function is disabled.

11.6.4.3 (ID 515) Frequency to Start DC Braking at Ramp Stop

Location in the menu: **P3.4.4.3**

Use this parameter to set the output frequency at which the DC-braking starts.

See [Illustration 63](#).

11.6.5 Flux Braking

11.6.5.1 (ID 520) Flux Braking

Location in the menu: **P3.4.5.1**

Use this parameter to enable Flux Braking. It is possible to use flux braking as an alternative to DC-braking. Flux braking increases the braking capacity in conditions where extra brake resistors are not necessary.

When braking is necessary, the system decreases the frequency and increases the flux in the motor. It increases the capacity of the motor to brake. The motor speed is controlled during braking.

NOTICE

Use the braking only intermittently. Flux braking converts energy into heat and can damage the motor.

11.6.5.2 (ID 519) Flux Braking Current

Location in the menu: **P3.4.5.2**

Use this parameter to set the current level for the flux braking.

11.7 I/O Configuration

11.7.1 Programming of Digital and Analog Inputs

11.7.1.1 General Information on the Programming

The programming of inputs of the AC drive is flexible. It is possible to use freely the available inputs of the standard and optional I/O for different functions.

It is possible to expand the available capacity of I/O with option boards. It is possible to install the option boards in the slots C, D, and E. For more data on the installation of option boards, see the Installation Guide.

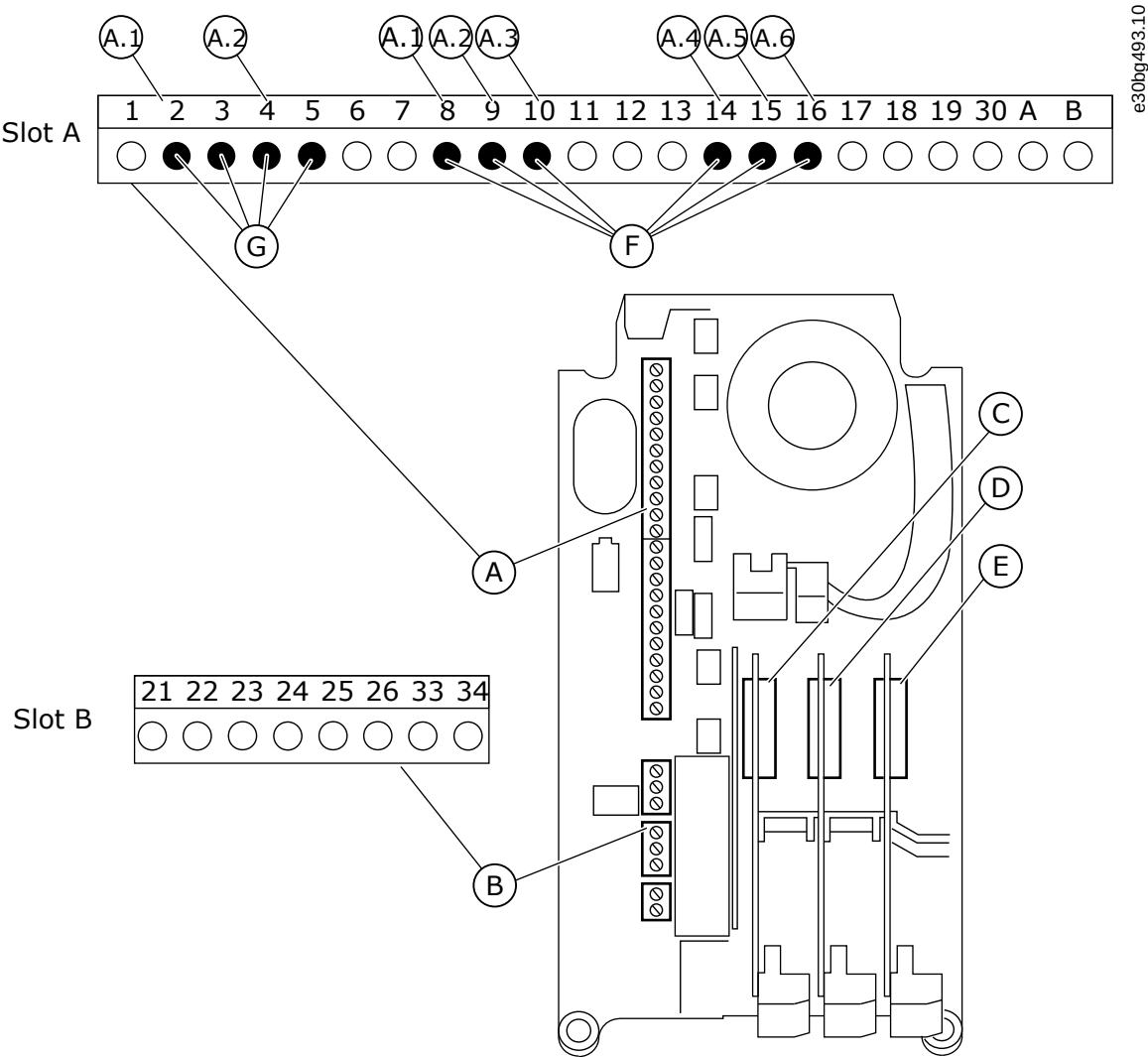


Illustration 51: The Option Board Slots and Programmable Inputs

A	Standard board slot A and its terminals	E	Option board slot E
B	Standard board slot B and its terminals	F	Programmable digital inputs (DI)
C	Option board slot C	G	Programmable analog inputs (AI)
D	Option board slot D		

11.7.1.2 Programming of Digital Inputs

Find the applicable functions for digital inputs as parameters in parameter group M3.5.1. To give a digital input to a function, set a value to the correct parameter. The list of applicable functions shows in [Table 44](#).

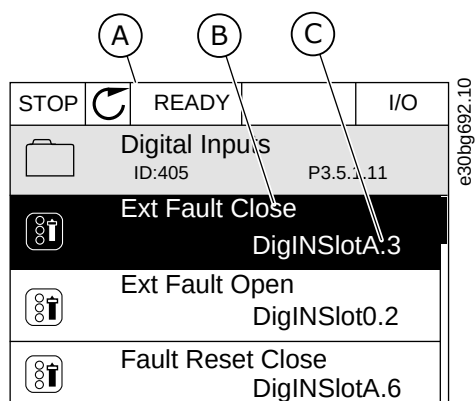


Illustration 52: The Digital Inputs Menu in the Graphical Display

A	The graphical display	C	The value of the parameter, that is, the set digital input
B	The name of the parameter, that is, the function		

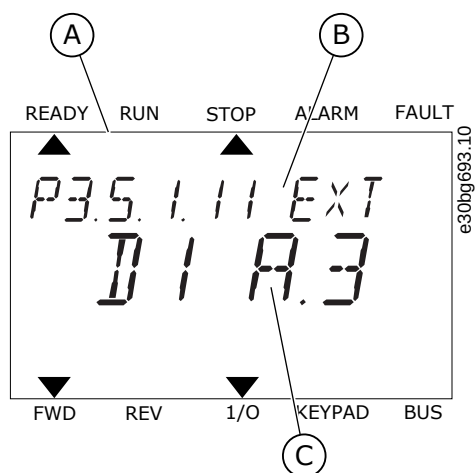


Illustration 53: The Digital Inputs Menu in the Text Display

A	The text display	C	The value of the parameter, that is, the set digital input
B	The name of the parameter, that is, the function		

Example

In the standard I/O board compilation, there are 6 digital inputs available: the slot A terminals 8, 9, 10, 14, 15 and 16.

Input type (graphical display)	Input type (text display)	Slot	Input #	Explanation
DigIN	dl	A	1	Digital input #1 (terminal 8) on a board in Slot A (standard I/O board).
DigIN	dl	A	2	Digital input #2 (terminal 9) on a board in Slot A (standard I/O board).
DigIN	dl	A	3	Digital input #3 (terminal 10) on a board in Slot A (standard I/O board).
DigIN	dl	A	4	Digital input #4 (terminal 14) on a board in Slot A (standard I/O board).
DigIN	dl	A	5	Digital input #5 (terminal 15) on a board in Slot A (standard I/O board).
DigIN	dl	A	6	Digital input #6 (terminal 16) on a board in Slot A (standard I/O board).

The function External Fault Close, the location of which is the menu M3.5.1, is parameter P3.5.1.11. It gets the default value DigIN SlotA.3 in the graphical display, and dI A.3 in the text display. After this selection, a digital signal to the digital input DI3 (terminal 10) controls External Fault Close.

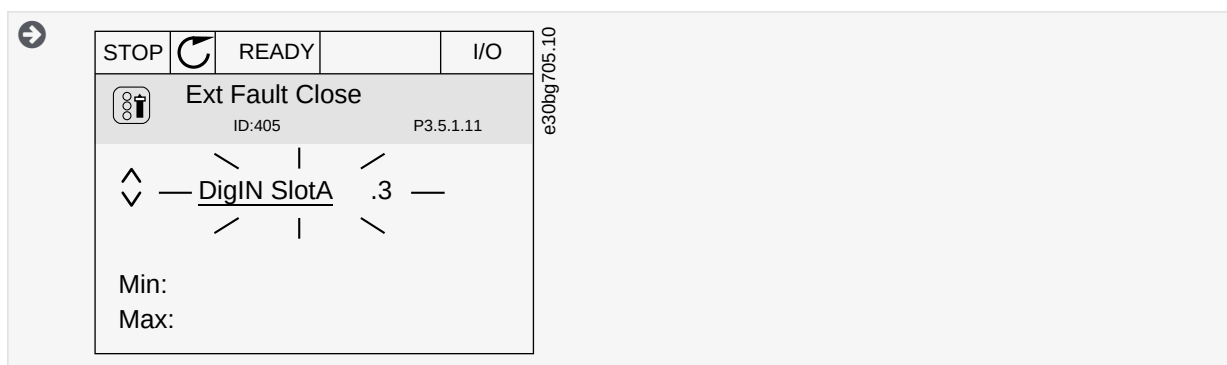
Index	Parameter	Default	ID	Description
P3.5.1.11	External fault close	DigIN SlotA.3	405	OPEN = OK CLOSED = External fault

To change the input from DI3 to, for example, DI6 (terminal 16) on the standard I/O, obey these instructions.

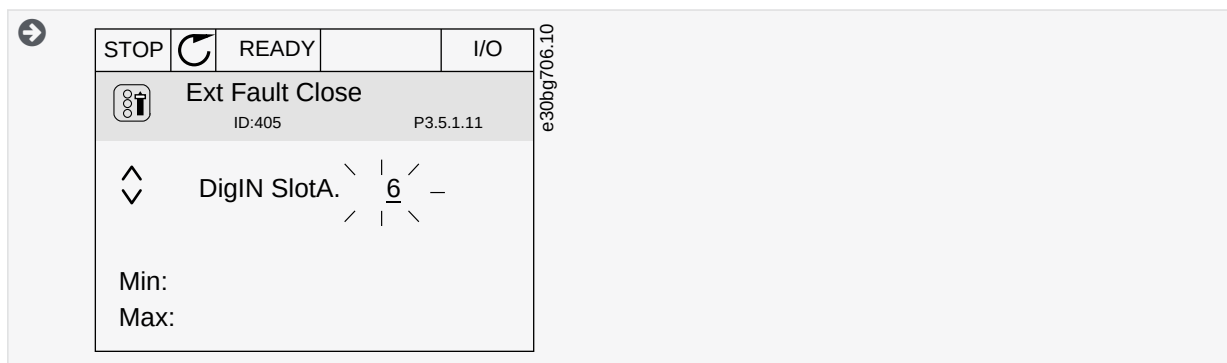
11.7.1.3 Programming of Digital Inputs in the Graphical Display

Procedure

1. Select a parameter. To go into the Edit mode, push the arrow button Right.
2. In the Edit mode, the slot value DigIN SlotA is underlined and blinks. If there are more digital inputs available in the I/O, for example, because of option boards in slots C, D, or E, select them.



3. To activate the terminal 3, push the arrow button Right again.
4. To change the terminal to 6, push the arrow button Up 3 times. Accept the change with the [OK] button.



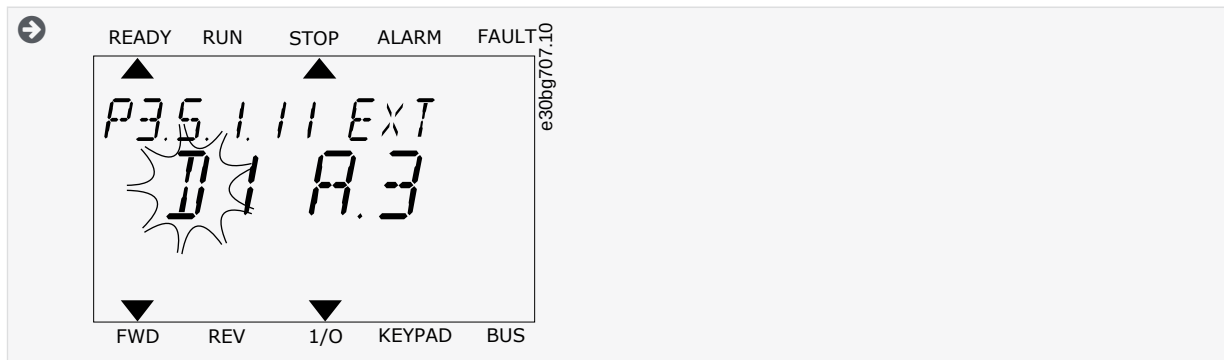
5. If the digital input DI6 was already used for some other function, a message shows on the display. Change one of these selections.

11.7.1.4 Programming of Digital Inputs in the Text Display

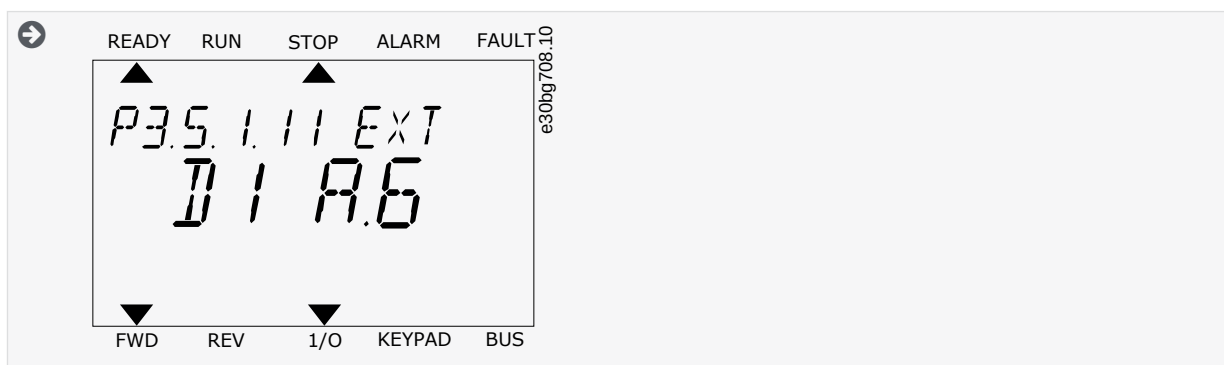
Procedure

1. Select a parameter. To go into the Edit mode, push the [OK] button.

- In the Edit mode, the letter D blinks. If there are more digital inputs available in the I/O, for example, because of option boards in slots C, D or E, select them.



- To activate the terminal 3, push the arrow button Right again. The letter D stops blinking.
- To change the terminal to 6, push the arrow button Up 3 times. Accept the change with the [OK] button.



- If the digital input DI6 was already used for some other function, a message scrolls on the display. Change one of these selections.

It is also possible to give time channels to digital inputs. There is more data about it in [Table 79](#).

After the steps, a digital signal to the digital input DI6 controls the function External Fault Close. The value of a function can be DigIN Slot0.1 (in the graphical display) or DI 0.1 (in the text display). In these conditions, a terminal was not given to the function, or the input was set to be always OPEN. It is the default value of most of parameters in the group M3.5.1. On the other hand, some inputs have the default value always CLOSED. Their value shows DigIN Slot0.2 in the graphical display and DI 0.2 in the text display.

11.7.1.5 Programming of Analog Inputs

It is possible to select the target input for the signal of the analog frequency reference from the available analog inputs.

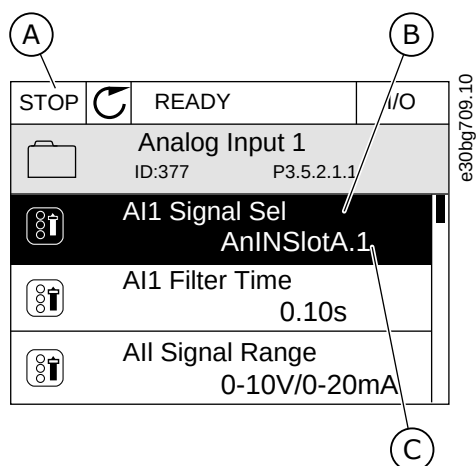


Illustration 54: The Analog Inputs Menu in the Graphical Display

A	The graphical display	C	The value of the parameter, that is, the set analog input
B	The name of the parameter		

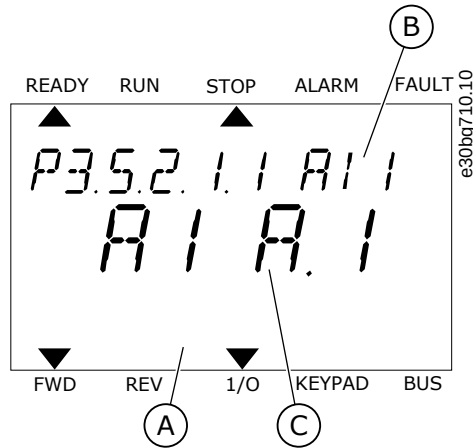


Illustration 55: The Analog Inputs Menu in the Text Display

A	The text display	C	The value of the parameter, that is, the set analog input
B	The name of the parameter		

Example

In the standard I/O board compilation, there are 2 analog inputs available: the slot A terminals 2/3 and 4/5.

Input type (graphical display)	Input type (text display)	Slot	Input #	Explanation
AnIN	AI	A	1	Analog input #1 (terminals 2/3) on a board in Slot A (standard I/O board).
AnIN	AI	A	2	Analog input #2 (terminals 4/5) on a board in Slot A (standard I/O board).

The location of the parameter P3.5.2.1.1 AI1 Signal Selection is the menu M3.5.2.1. The parameter gets the default value AnIN SlotA.1 in the graphical display or AI A.1 in the text display. The target input for the signal of the analog frequency reference AI1 is then the analog input in the terminals 2/3. Use the DIP switches to set the signal to be voltage or current. See the Installation manual for more data.

Index	Parameter	Default	ID	Description
P3.5.2.1.1	AI1 Signal Selection	AnIN SlotA.1	377	

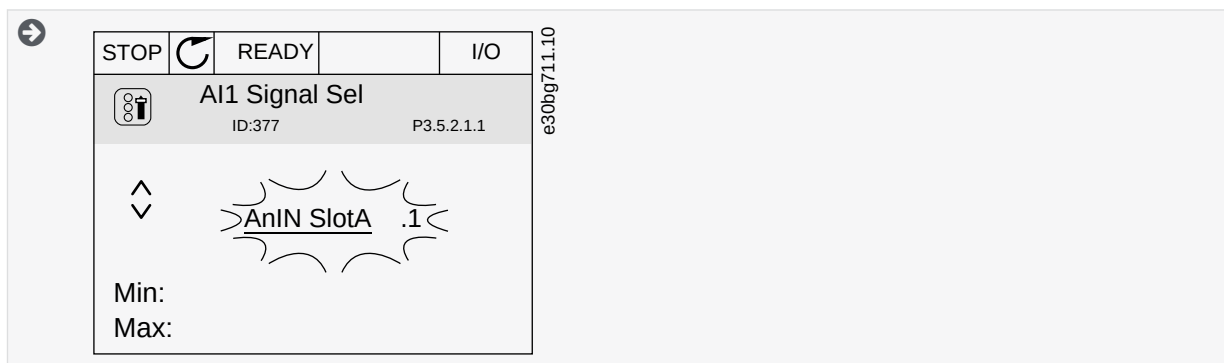
To change the input from AI1 to, for example, the analog input on your option board in slot C, obey these instructions.

11.7.1.6 Programming of Analog Inputs in the Graphical Display

Procedure

1. To select the parameter, push the arrow button Right.

- In the Edit mode, the value AnIN SlotA is underlined and blinks.

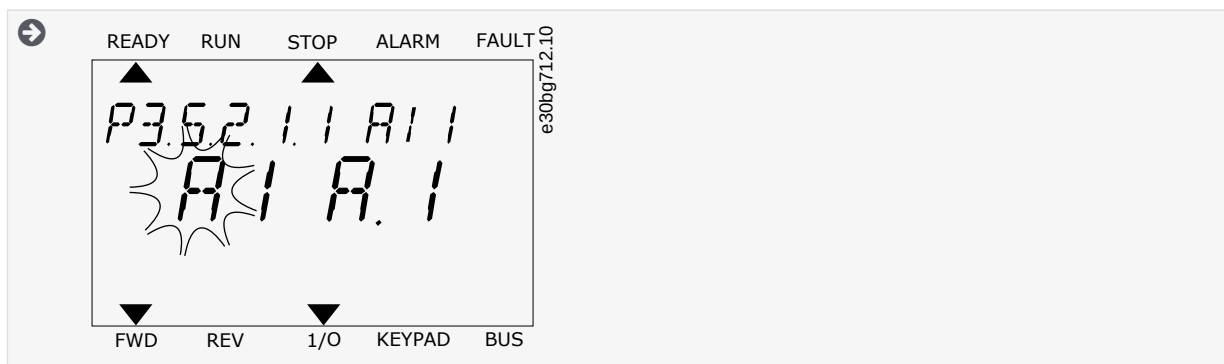


- To change the value to AnIN SlotC, push the arrow button Up. Accept the change with the [OK] button.

11.7.1.7 Programming of Analog Inputs in the Text Display

Procedure

- To select the parameter, push the [OK] button.
- In the Edit mode, the letter A blinks.



- To change the value to C, push the arrow button Up. Accept the change with the [OK] button.

11.7.2 Descriptions of Signal Sources

Table 130: Descriptions of Signal Sources

Source	Function
Slot0.#	Digital inputs: Use this function to set a digital signal to be in a constant OPEN or CLOSED state. The manufacturer set some signals so that they are always in the CLOSED state, for example parameter P3.5.1.15 (Run Enable). The Run Enable signal is always on if it is not changed. # = 1: Always OPEN # = 2–10: Always CLOSED Analog inputs (used for testing purposes): # = 1: Analog input = 0% of the signal strength # = 2: Analog input = 20% of the signal strength # = 3: Analog input = 30% of the signal strength, and so on # = 10: Analog input = 100% of the signal strength
SlotA.#	Number (#) agrees to a digital input in slot A.
SlotB.#	Number (#) agrees to a digital input in slot B.
SlotC.#	Number (#) agrees to a digital input in slot C.
SlotD.#	Number (#) agrees to a digital input in slot D.

Source	Function
SlotE.#	Number (#) agrees to a digital input in slot E.
TimeChannel.#	1=Time Channel1, 2=Time Channel2, 3=Time Channel3
FieldbusCW.#	Number (#) refers to a control word bit number.
FieldbusPD.#	Number (#) refers to the process data 1 bit number.

11.7.3 Default Functions of Programmable Inputs

Table 131: Default Functions of the Programmable Digital and Analog Inputs

Input	Terminal(s)	Reference	Function	Parameter index
DI1	8	A.1	Control Signal 1 A	P3.5.1.1
DI2	9	A.2	Control Signal 2 A	P3.5.1.2
DI3	10	A.3	External Fault Close	P3.5.1.11
DI4	14	A.4	Preset Frequency Selection 0	P3.5.1.21
DI5	15	A.5	Preset Frequency Selection 1	P3.5.1.22
DI6	16	A.6	Fault Reset Close	P3.5.1.13
AI1	2/3	A.1	AI1 Signal Selection	P3.5.2.1.1
AI2	4/5	A.2	AI2 Signal Selection	P3.5.2.2.1

11.7.4 Digital Inputs

The parameters are functions that can be connected to a digital input terminal. The text DigIn Slot A.2 means the second input on the slot A. It is also possible to connect the functions to time channels. The time channels work as terminals.

It is possible to monitor the statuses of the digital inputs and the digital outputs in the Multimonitoring view.

11.7.4.1 (ID 403) Control Signal 1 A

Location in the menu: **P3.5.1.1**

Use this parameter to select the digital input signal (Control Signal 1) that starts and stops the drive when the control place is I/O A (FWD).

11.7.4.2 (ID 404) Control Signal 2 A

Location in the menu: **P3.5.1.2**

Use this parameter to select the digital input signal (Control Signal 2) that starts and stops the drive when the control place is I/O A (REV).

11.7.4.3 (ID 434) Control Signal 3 A

Location in the menu: **P3.5.1.3**

Use this parameter to select the digital input signal (Control Signal 3) that starts and stops the drive when the control place is I/O A.

11.7.4.4 (ID 423) Control Signal 1 B

Location in the menu: **P3.5.1.4**

Use this parameter to select the digital input signal (Control Signal 1) that starts and stops the drive when the control place is I/O B.

11.7.4.5 (ID 424) Control Signal 2 B

Location in the menu: **P3.5.1.5**

Use this parameter to select the digital input signal (Control Signal 2) that starts and stops the drive when the control place is I/O B.

11.7.4.6 (ID 435) Control Signal 3 B

Location in the menu: **P3.5.1.6**

Use this parameter to select the digital input signal (Control Signal 3) that starts and stops the drive when the control place is I/O B.

11.7.4.7 (ID 425) I/O B Control Force

Location in the menu: **P3.5.1.7**

Use this parameter to select the digital input signal that switches the control place from I/O A to I/O B.

11.7.4.8 (ID 343) I/O B Reference Force

Location in the menu: **P3.5.1.8**

Use this parameter to select the digital input signal that switches the frequency reference source from I/O A to I/O B.

11.7.4.9 (ID 411) Fieldbus Control Force

Location in the menu: **P3.5.1.9**

Use this parameter to select the digital input signal that switches the control place and the frequency reference source to Fieldbus (from I/O A, I/O B or Local control).

11.7.4.10 (ID 410) Keypad Control Force

Location in the menu: **P3.5.1.10**

Use this parameter to select the digital input signal that switches the control place and the frequency reference source to Keypad (from any control place).

11.7.4.11 (ID 405) External Fault Close

Location in the menu: **P3.5.1.11**

Use this parameter to select the digital input signal that activates an external fault.

11.7.4.12 (ID 406) External Fault Open

Location in the menu: **P3.5.1.12**

Use this parameter to select the digital input signal that activates an external fault.

11.7.4.13 (ID 414) Fault Reset Close

Location in the menu: **P3.5.1.13**

Use this parameter to select the digital input signal that resets all active faults. Active faults are reset when the state of the digital input changes from open to closed (rising edge).

11.7.4.14 (ID 213) Fault Reset Open

Location in the menu: **P3.5.1.14**

Use this parameter to select the digital input signal that resets all active faults. Active faults are reset when the state of the digital input changes from closed to open (falling edge).

11.7.4.15 (ID 407) Run Enable

Location in the menu: **P3.5.1.15**

Use this parameter to select the digital input signal that sets the drive to Ready state. When the contact is OPEN, the start of the motor is disabled. When the contact is CLOSED, the start of the motor is enabled.

The state of the drive remains in 'Not Ready' if the state of this signal is 'opened'. If the Run Enable signal is used for stopping the drive, the drive will always coast to stop regardless of the selection in parameter P3.2.5 Stop Function.

11.7.4.16 (ID 1041) Run Interlock 1

Location in the menu: **P3.5.1.16**

Use this parameter to select the digital input signal that prevents to start the drive. The drive can be ready but start is not possible when the state of the interlock signal is 'open' (damper interlock).

11.7.4.17 (ID 1042) Run Interlock 2

Location in the menu: **P3.5.1.17**

Use this parameter to select the digital input signal that prevents starting the drive. The drive can be ready but start is not possible when the state of the interlock signal is 'open' (damper interlock).

If an interlock is active, the drive cannot start.

It is possible to use this function to prevent the start of the drive when the damper is closed. If an interlock is activated during the operation of the drive, the drive stops.

11.7.4.18 (ID 1044) Motor Preheat ON

Location in the menu: **P3.5.1.18**

Use this parameter to select the digital input signal that activates Motor Preheat function. The Motor Preheat function feeds DC-current to the motor when the drive is in the stop state.

11.7.4.19 (ID 408) Ramp 2 Selection

Location in the menu: **P3.5.1.19**

Use this parameter to select the digital input signal that selects the ramp time to be used.

11.7.4.20 (ID 415) Acc/Dec Prohibit

Location in the menu: **P3.5.1.20**

Use this parameter to select the digital input signal that prevents the acceleration and the deceleration of the drive. No acceleration or deceleration is possible until the contact is open.

11.7.4.21 (ID 419) Preset Frequency Selection 0

Location in the menu: **P3.5.1.21**

Use this parameter to set the digital input signal that selects the preset frequencies.

11.7.4.22 (ID 420) Preset Frequency Selection 1

Location in the menu: **P3.5.1.22**

Use this parameter to set the digital input signal that selects the preset frequencies.

11.7.4.23 (ID 421) Preset Frequency Selection 2

Location in the menu: **P3.5.1.23**

Use this parameter to set the digital input signal that selects the preset frequencies.

11.7.4.24 (ID 418) Motor Potentiometer Up

Location in the menu: **P3.5.1.24**

Use this parameter to increase the output frequency with a digital input signal. The motor potentiometer reference INCREASES until the contact is open.

11.7.4.25 (ID 417) Motor Potentiometer Down

Location in the menu: **P3.5.1.25**

Use this parameter to decrease the output frequency with a digital input signal. The motor potentiometer reference DECREASES until the contact is open.

11.7.4.26 (ID 1213) Quick Stop Activation

Location in the menu: **P3.5.1.26**

Use this parameter to select the digital input signal that activates a Quick Stop function. The Quick Stop function stops the drive regardless of the control place or the state of the control signals.

11.7.4.27 (ID 447) Timer 1

Location in the menu: **P3.5.1.27**

Use this parameter to select the digital input signal that starts the timer. The timer starts when this signal is deactivated (falling edge). The output is deactivated when the time defined in the duration parameter has elapsed.

11.7.4.28 (ID 448) Timer 2

Location in the menu: **P3.5.1.28**

Use this parameter to select the digital input signal that starts the timer. The timer starts when this signal is deactivated (falling edge). The output is deactivated when the time defined in the duration parameter has elapsed.

11.7.4.29 (ID 449) Timer 3

Location in the menu: **P3.5.1.29**

Use this parameter to select the digital input signal that starts the timer. The timer starts when this signal is deactivated (falling edge). The output is deactivated when the time defined in the duration parameter has elapsed.

11.7.4.30 (ID 1046) PID Setpoint Boost

Location in the menu: **P3.5.1.30**

Use this parameter to select the digital input signal that activates the boost for the PID setpoint value. The timer starts when this signal is deactivated (falling edge). The output is deactivated when the time defined in the duration parameter has elapsed.

11.7.4.31 (ID 1047) PID Setpoint Selection

Location in the menu: **P3.5.1.31**

Use this parameter to set the digital input signal that selects the PID setpoint value to be used.

11.7.4.32 (ID 1049) External PID Start Signal

Location in the menu: **P3.5.1.32**

Use this parameter to select the digital input signal that starts and stops the external PID controller.

NOTICE

This parameter has no effect if the external PID controller is not enabled in Group 3.14.

11.7.4.33 (ID 1048) External PID Setpoint Selection

Location in the menu: **P3.5.1.33**

Use this parameter to set the digital input signal that selects the PID setpoint value to be used.

11.7.4.34 (ID 490) Reset Maintenance Counter

Location in the menu: **P3.5.1.34**

Use this parameter to select the digital input that reset the value of the Maintenance Counter.

11.7.4.35 (ID 530) Flushing Reference Activation

Location in the menu: **P3.5.1.36**

Use this parameter to select the digital input signal that activates the Flushing function. The flushing frequency reference is bidirectional and a reverse command does not affect the direction of the flushing reference.

NOTICE

If the input is activated, the drive starts.

11.7.4.36 (ID 1596) Fire Mode Activation OPEN

Location in the menu: **P3.5.1.38**

Use this parameter to select the digital input signal that activates the Fire Mode function. This parameter activates the Fire Mode if it is enabled with a correct password.

11.7.4.37 (ID 1619) Fire Mode Activation CLOSE

Location in the menu: **P3.5.1.39**

Use this parameter to select the digital input signal that activates the Fire Mode function. This parameter activates the Fire Mode if it is enabled with a correct password.